MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Mansourah Massarah Gold Project
Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: DESIGN CALCULATION
49FE FUEL STORAGE FACILITY:
COMPRESSOR ROOM & VAPORIZER ROOM
DESIGN OF BUILDING & FOUNDATION

Document Number MD-216-49FE-EG-CV-CAL-1627
Revision: D01
Revision Date: 26.12.2023
Status: AS BUILT

Outotec Document Number:	900707	GAU01	AG50	CED01	49001	OU100703981	1
L&T Document Number:	O19045	080	C	K	24312	001	1

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	20.10.2020	FIRST ISSUE	TEC	AVG	SJM	
C01	20.11.2020	ISSUED FOR CONSTRUCTION	TEC	AVG	SJM	
D01	26.12.2023	AS BUILT	TEC	AVG	SJM	

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 4
1. INTENT 5
2. EXECUTIVE SUMMARY 5
3. SCOPE 5
4. DESIGN METHODOLOGY & ASSUMPTIONS 5
4.1. Codes 6
4.2. Geotechnical investigations 6
4.3. Material Properties 6
5. DESIGN LOADS 7
5.1. Dead Load(DL) 7
5.1.1. Roof Dead Loads 7
5.1.2. Ground Floor Walls 9
5.1.3. Roof Live Loads (Lr) 9
5.2. Wind Load (WL) 9
5.3. Seismic Load(E) 10
5.4. Thermal/Temperature Loads(Self-restraining ,Thermal Expansion & Contraction Loads) 11
6. DESIGN COMBINATIONS 11
7. CRACKED SECTIONS 12
8. DEFLECTIONS 12
8.1. Maximum Allowable Deflection 12
9. CONCRETE ROOF BEAM DESIGN 15
9.1. Design of Ledger Beam(LB1) 15
9.1.1. Staad Output for LB1 18
9.2. Design of Beam(B1) 19
9.2.1. Staad Output for B1 21
10. CONCRETE GRADE SLAB DESIGN 22
11. DESIGN OF GRADE BEAM 23

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

11.1. Grade Beam(GB1) 23
11.1.1. Staad Output for GB1 26
12. CONCRETE COLUMN DESIGN 27
12.1. Design of Column-(C1)_Vaporizer Room 27
12.1.1. Staad Output for Column C1_Vaporizer Room 27
12.2. Design of Column-(C2)_Vaporizer Room 28
12.2.1. Staad Output for Column C2_Vaporizer Room 29
12.3. Design of Column-(C3)_Vaporizer Room 30
12.3.1. Staad Output for Column C3_Vaporizer Room 31
12.4. Design of Column-(C1)_Compressor Room 31
12.4.1. Staad Output for Column C1_Compressor Room 32
12.5. Design of Column-(C2)_Compressor Room 33
12.5.1. Staad Output for Column C2_Compressor Room 34
13. FOUNDATION DESIGN 35
13.1. FOUNDATION DESIGN CALCULATIONS_ Vaporizer Room 36
13.2. FOUNDATION DESIGN CALCULATIONS_ COMPRESSOR ROOM 59
References 71
ATTACHMENT 1 STAAD INPUT DATA_ VAPORIZER ROOM 72
ATTACHMENT 2 STAAD INPUT DATA_ COMPRESSOR ROOM 78
ATTACHMENT 3 LAYOUT DRAWING 84

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Vaporizer Room and Compressor Room in Fuel Storage area.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Vaporizer Room and Compressor Room only.

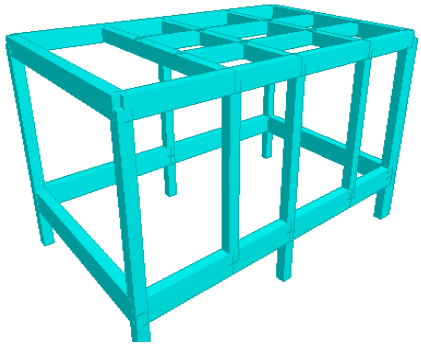
4. DESIGN METHODOLOGY & ASSUMPTIONS

These Buildings are Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab). The Superstructure for this Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ and Steel Columns & Beams) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base that are assumed to be fixed. The Foundation design has been done using excel by importing Column Base Reactions from analysis Software.

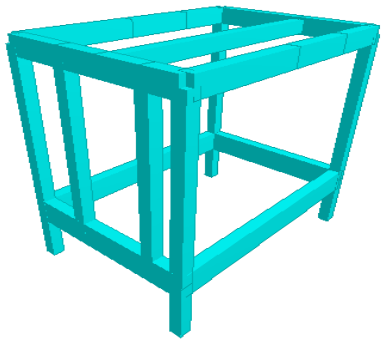
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



STAAD 3D MODEL _Vaporizer Room



STAAD 3D MODEL _Compressor Room

4.1. Codes

[1] SBC-301-2007- Structural Loading & Forces.

[2] SBC-303-2007- Structural Soil & Foundations.

[3] SBC-304-2007- Structural Concrete Structures.

[4] SASO SSA 2 - Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 - Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 - Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 - Building Code Requirements.

[8] PCI –MNL-120-04 - Design Handbook, 6th edition.

[9] SBC-306-2007- Structural – Steel Structures.

4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5. DESIGN LOADS

5.1. Dead Load(DL)

5.1.1. Roof Dead Loads

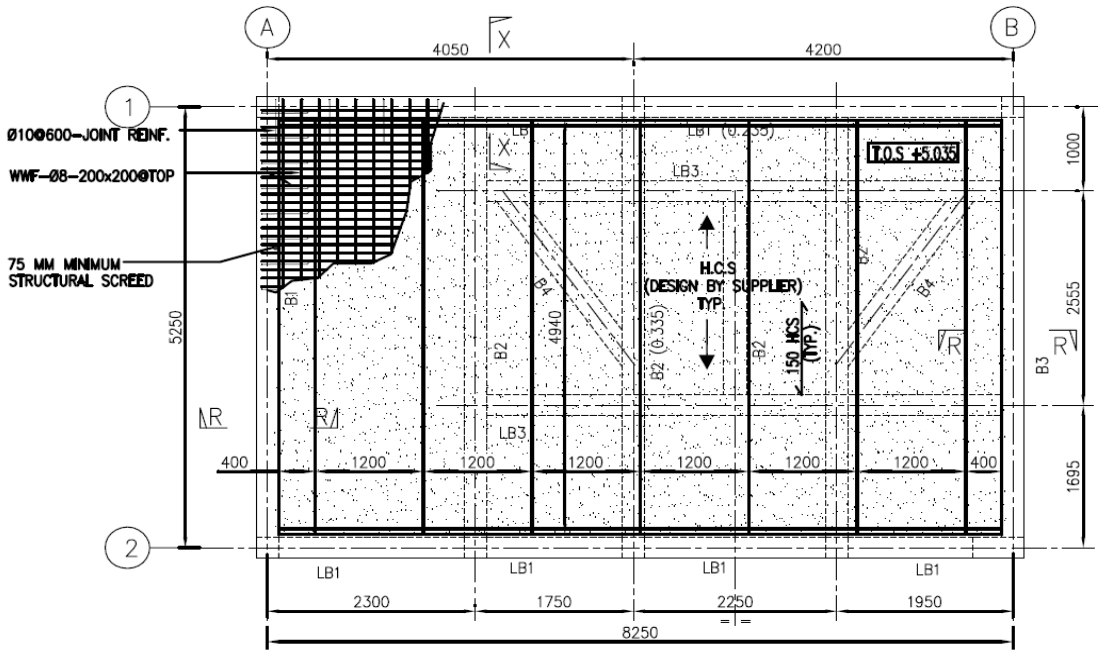
- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = **0.2 kN/m²**



ROOF HOLLOW CORE SLAB LAYOUT
SCALE 1:50

- Hollow Core Slab

(HCS-150)(To be confirmed by Vendor) b/w Grid Line's (1 & 2) = **2.52 KN/m2**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Prestressed Hollowcore Floor										
Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m^2 for Finishes Characteristic Service Loads Kn/m^2							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/ m²**

Therefore

Total Dead Load =3.1+0.2+2.52= 5.82 **kN/m²**

TOTAL DL as UDL ON BEAMS

- UDL on LB's Along Grid Line-1 & 2= 5.25 x 5.82/2 = **15.3 Kn/m** (For Vaporizer Room)
- UDL on LB's Along Grid Line-1 & 2= 3.65 x 5.82/2 = **10.7 Kn/m** For Compressor Room)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5.1.2. Ground Floor Walls

- 250 thk. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x 4.30m clear height = **17.2 kN/m**

5.1.3. Roof Live Loads (Lr)

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Ordinary Flat (Inaccessible) = **0.5 kN/m²**

TOTAL Lr as UDL ON BEAMS

- UDL on LB's Along Grid Line = 5.25 x 0.50/2 = **1.3125 Kn/m** (For Vaporizer Room)
- UDL on LB's Along Grid Line = 3.65 x 0.50/2 = **0.9125 Kn/m** (For Compressor Room)

5.2. Wind Load (WL)

Wind loads were automatically generated in STAAD as per ASCE-07. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON-PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

The Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

	Vaporizer Room	Compressor Room
Type of roof	Flat	Flat
Length of building (Z - direction)	8500 mm	5500 mm
Width of building (X - direction)	5500 mm	3900 mm
Height to eaves	5535 mm	5535 mm
Mean height	5535 mm	5535 mm

General wind load requirements:

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Basic wind speed:	V = 49 m/s (or) 177 kmph (Refer Clause-4.2.6, Doc-MD-216-1000-EG-ST-CRT-1001)
Risk category:	II (As per SBC-301, Table 1.6-1)
Velocity pressure coefficient:	K_d = **0.85** (As per SBC-301, Table 6.4-1)
Exposure category:	D (As per SBC-301, Section 6.4.2.3)
Enclosure classification:	Enclosed buildings
Gust effect factor:	G_f = **0.85** (As per SBC-301, Section-7.2.7.1)
Importance factor:	1 (As per SBC-301, Section-9.5)
Topography factor:	Kzt = 1.0 (As per SBC-301, Section 6.4.3.2)

5.3. Seismic Load(E)

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*Ss = 2/3 * 1.6 * 0.1 = 0.106 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 Sec

S_{DS} Damped Design Spectral Response Acceleration@0.2 Sec

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.4. Thermal/Temperature Loads(Self-restraining ,Thermal Expansion & Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

W = Wind Load

E = Seismic Load

T = Temperature Load

7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

8. DEFLECTIONS

8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

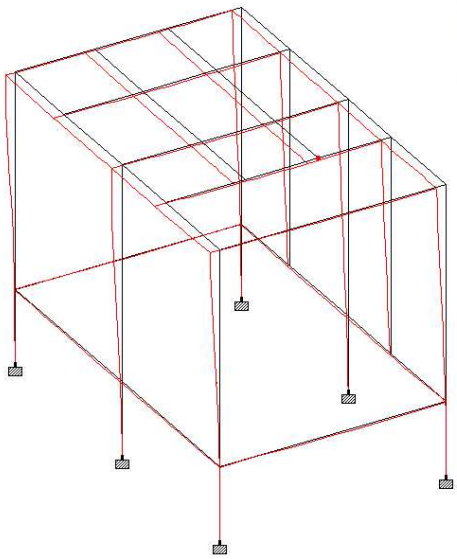
L/600: for members supporting masonry under total loads

Where, L = span of member between support

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



Veporizer Room - Node Displacements:

All | Summary

	Node	L/C	Horizontal X mm	Vertical Y mm	Horizontal Z mm	Resultant mm	Rotational rX rad	Rotational rY rad
Max X	27	18 0.6D+WX	**18.496**	-1.687	1.148	18.608	-0.000	0.000
Min X	31	45 D+WX2	**-22.120**	-3.890	0.942	22.479	-0.000	-0.000
Max Y	21	42 0.6D+WZ	1.133	**0.074**	-1.149	1.615	0.000	0.000
Min Y	26	23 D+WZ1	-2.672	**-4.066**	15.177	15.938	0.000	0.001
Max Z	5	23 D+WZ1	-4.461	-0.347	**16.802**	17.388	0.002	0.001
Min Z	7	42 0.6D+WZ	-4.439	-0.167	**-16.322**	16.916	-0.001	-0.001
Max rX	5	23 D+WZ1	-4.461	-0.347	16.802	17.388	**0.002**	0.001
Min rX	7	46 D+WZ2	-5.011	-0.300	-15.898	16.671	**-0.002**	-0.001
Max rY	32	19 0.6D+WZ	-3.785	-1.121	12.978	13.565	0.001	**0.001**
Min rY	32	40 D+0.75W	2.991	-1.562	-6.265	7.116	0.001	**-0.001**
Max rZ	16	45 D+WX2	-3.990	-0.233	0.188	4.001	-0.000	0.000
Min rZ	16	18 0.6D+WX	3.703	-0.151	0.070	3.706	-0.000	0.000
Max Rs	31	45 D+WX2	-22.120	-3.890	0.942	**22.479**	-0.000	-0.000

Maximum Deflection_ Vaporization Room

Maximum attained Drift as per Analysis from above figure's = **22.5 mm**

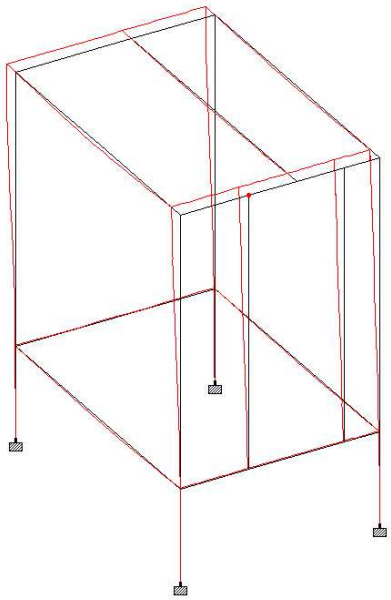
Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 5035/100 = **50.35 mm**

Hence the Structure is safe considering drift check…

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



Compressor Room - Node Displacements:

All / Summary

	Node	L/C	Horizontal X mm	Vertical Y mm	Horizontal Z mm	Resultant mm	Rotational rX rad	rY rad	rZ rad
Max X	5	18 0.6D+WX	**14.813**	-0.167	-2.479	15.020	0.001	-0.001	-0.001
Min X	6	45 D+WX2	**-15.139**	-0.315	-3.182	15.473	0.002	0.001	0.001
Max Y	1	10 DL	0.000	**0.000**	0.000	0.000	0.000	0.000	0.000
Min Y	34	40 D+0.75W	0.567	**-5.041**	-14.718	15.567	-0.000	0.000	-0.000
Max Z	5	19 0.6D+WZ	-0.517	-0.166	**17.639**	17.647	0.002	-0.000	-0.000
Min Z	39	46 D+WZ2	-0.269	-0.563	**-21.444**	21.453	-0.003	0.000	-0.000
Max rX	5	17 D+0.75W	0.098	-0.327	13.686	13.690	**0.003**	0.000	-0.000
Min rX	30	46 D+WZ2	-0.272	-0.300	-20.898	20.902	**-0.003**	0.001	-0.000
Max rY	34	39 D+0.75W	-8.215	-4.819	0.828	9.560	-0.000	**0.001**	-0.000
Min rY	35	18 0.6D+WX	12.552	-2.686	0.318	12.840	-0.000	**-0.001**	-0.000
Max rZ	27	45 D+WX2	-1.307	-0.161	-0.078	1.319	0.001	-0.000	**0.001**
Min rZ	12	22 D+WX1	1.292	-0.161	-0.101	1.306	0.001	0.000	**-0.001**
Max Rs	34	46 D+WZ2	-1.056	-4.912	-20.894	**21.489**	-0.000	-0.000	-0.000

Maximum Deflection_ Compressor Room

Maximum attained Drift as per Analysis from above figure's = **21.5 mm**

Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 5035/100 = **50.35 mm**

Hence the Structure is safe considering drift check…

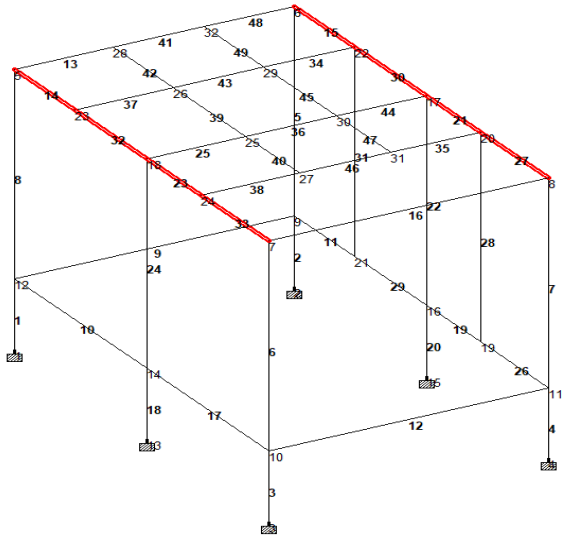
PROPRIETARY & CONFIDENTIAL

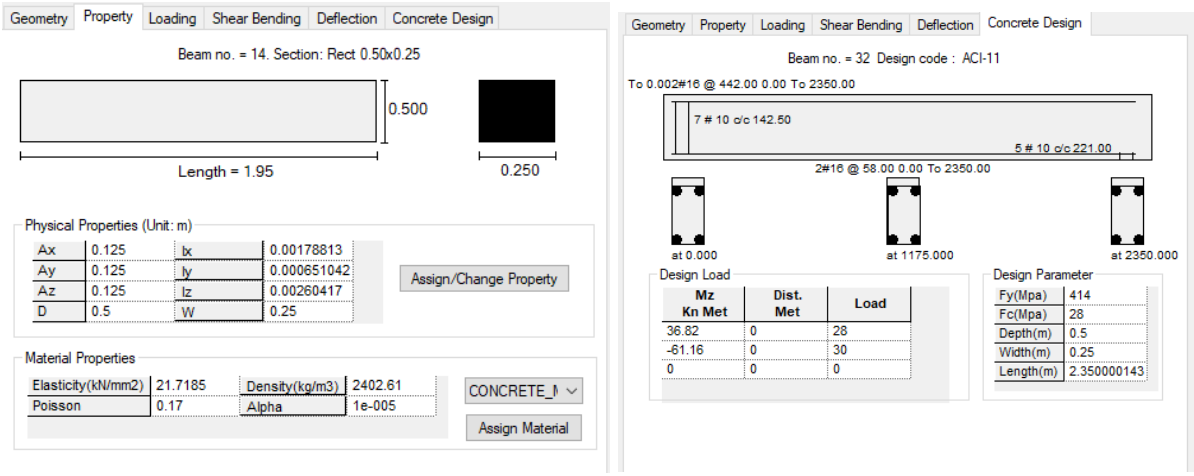
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9. CONCRETE ROOF BEAM DESIGN

9.1. Design of Ledger Beam(LB1)



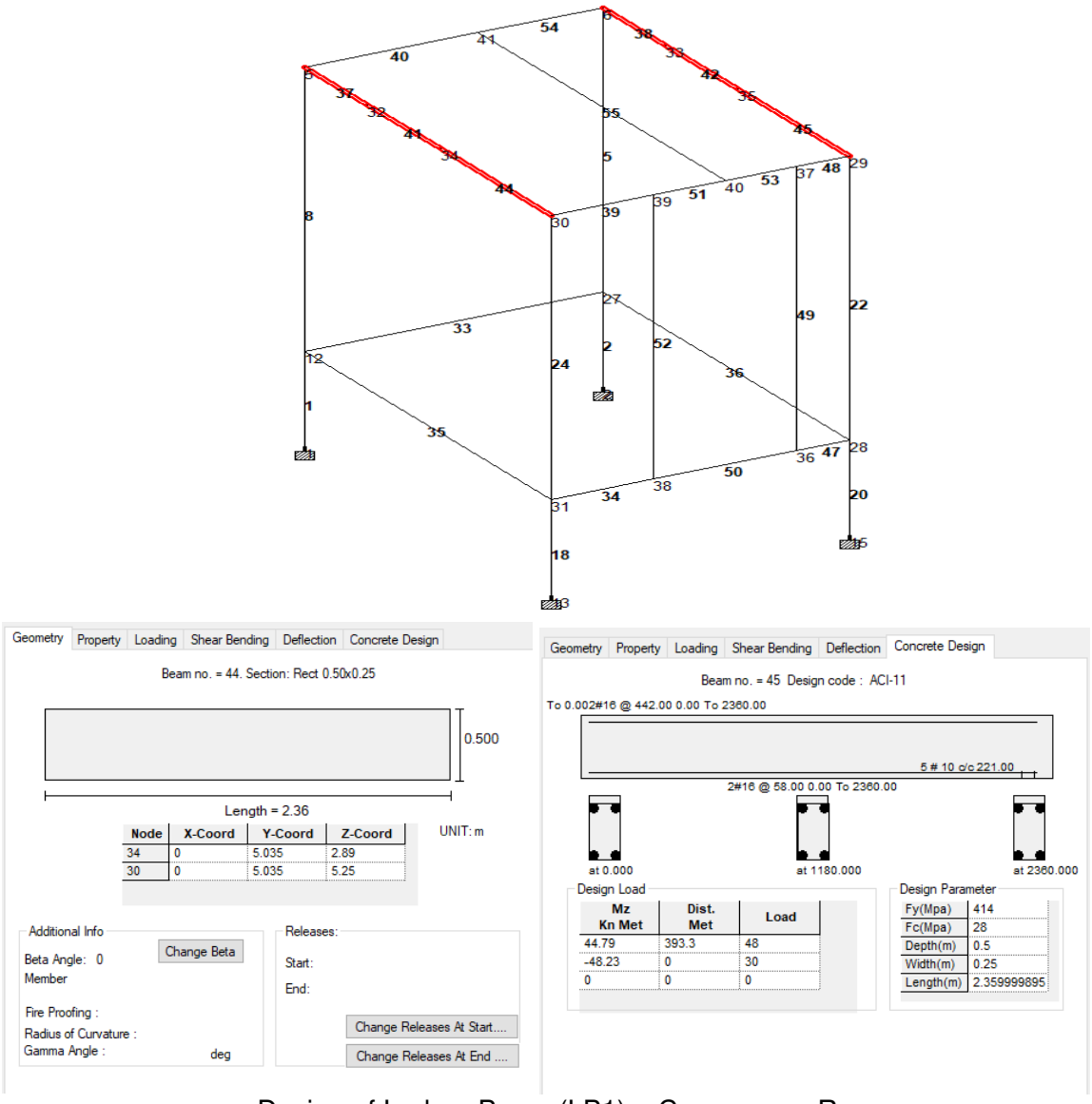


Design of Ledger Beam (LB1) _ Vaporizer Room

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



Design of Ledger Beam (LB1) _ Compressor Room

Maximum Moment required from Analysis = 62 **Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-16/2=442mm, Fc'=35 Mpa & ϕ =0.9

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Ast.req = **390 mm²**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm²**

Maximum Moment required from Analysis = **45 Knm (Sagging)**

Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

Ast.pro= **600mm² >>>>>>(Ast.min, Ast.req)**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

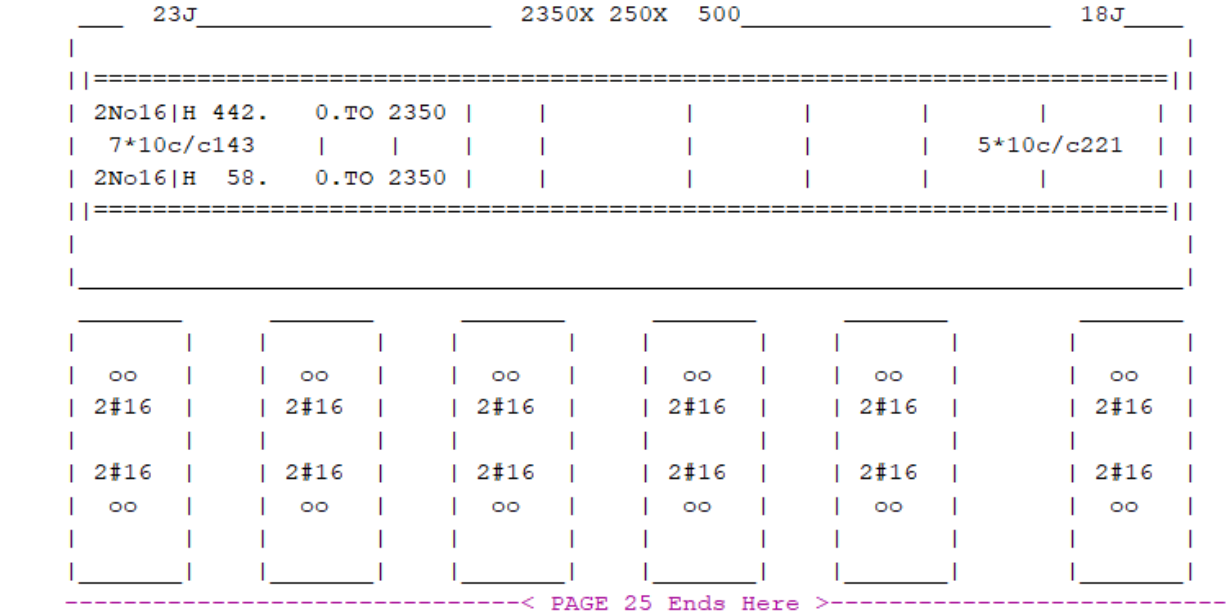
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.1.1. Staad Output for LB1

```
        B E A M  N O.    32 D E S I G N  R E S U L T S - SHEAR


   AT START SUPPORT - Vu=   17.35 KNS  Vc=   96.89 KNS  Vs=    0.00 KNS
   Tu=    3.43 KN-MET  Tc=    3.4 KN-MET  Ts=    4.6 KN-MET  LOAD    29
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR  735. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.13 SQ.CM.

   AT END   SUPPORT - Vu=   41.06 KNS  Vc=   97.14 KNS  Vs=    0.00 KNS
   Tu=    1.31 KN-MET  Tc=    3.4 KN-MET  Ts=    0.0 KN-MET  LOAD     3
   NO STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR  735. MM
```




```
-------------------------------< PAGE 25 Ends Here >--------------------------
   STAAD SPACE                                              -- PAGE NO.   26
```

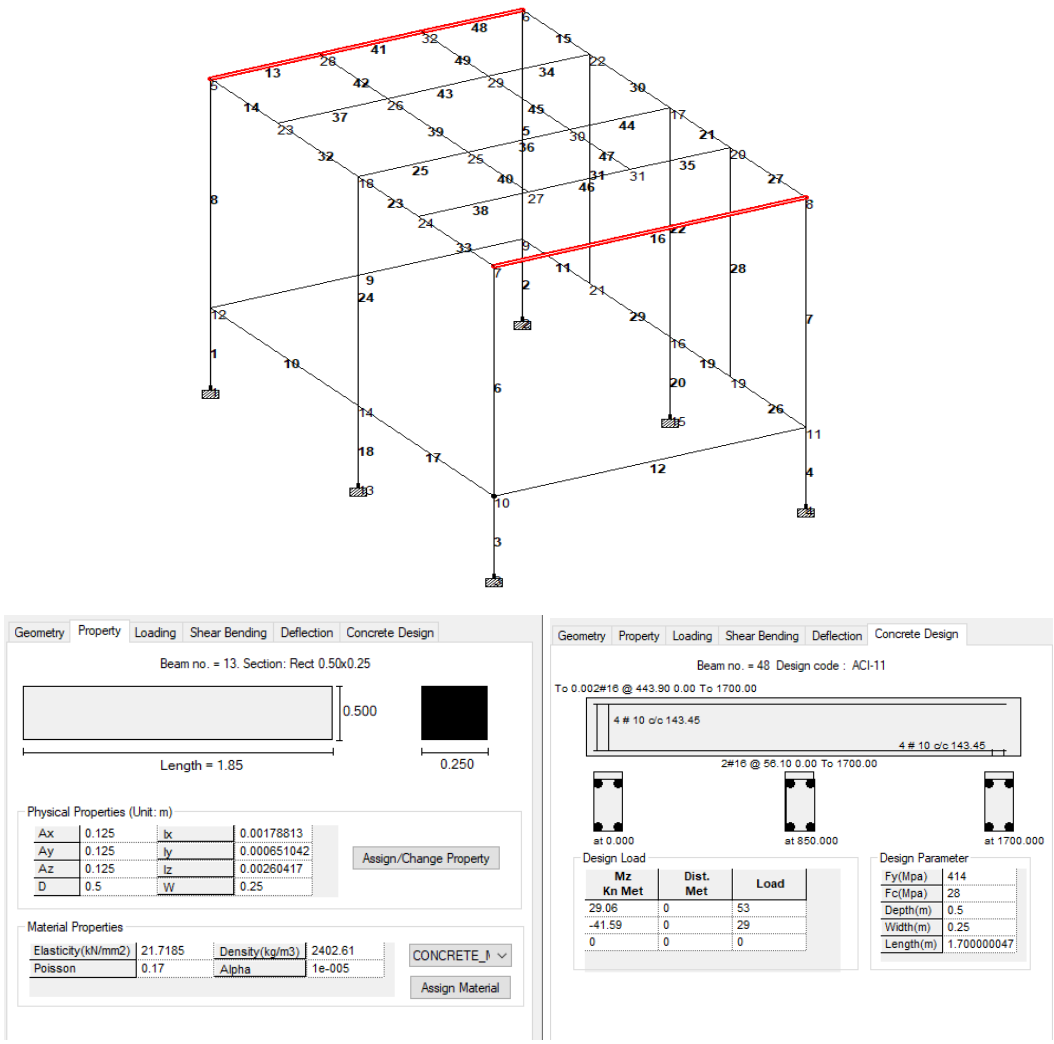
Note: Above result for shear check is for most critical beam in vaporizer Room. Same shear reinforcement detail has also been adopted for Compressor Room.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.2. Design of Beam(B1)



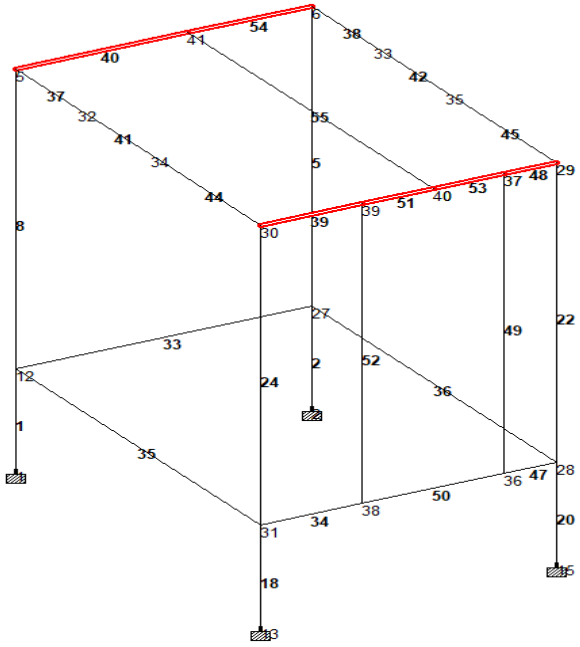
Design of Beam (B1) _ Vaporizer Room

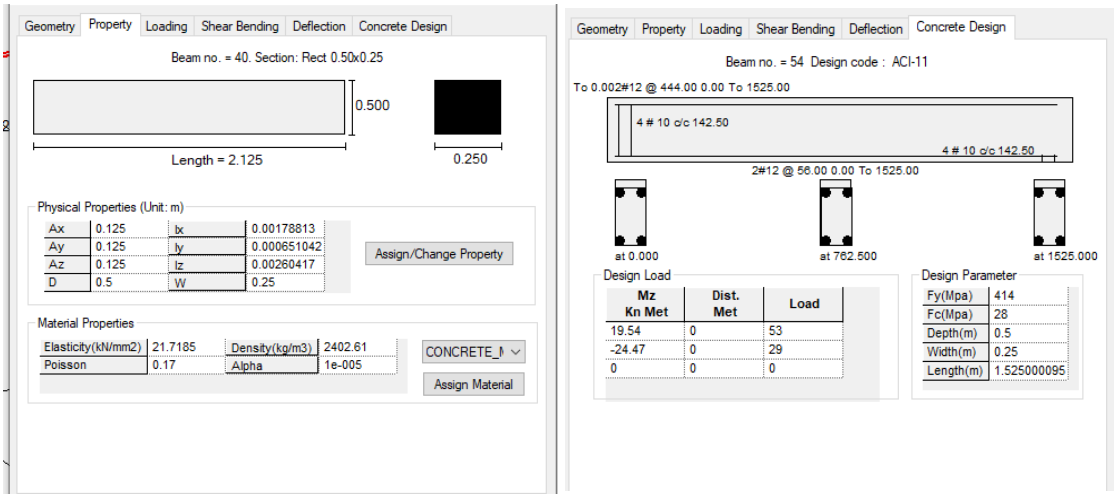
Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.





Design of Beam (B1) _ Compressor Room

Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

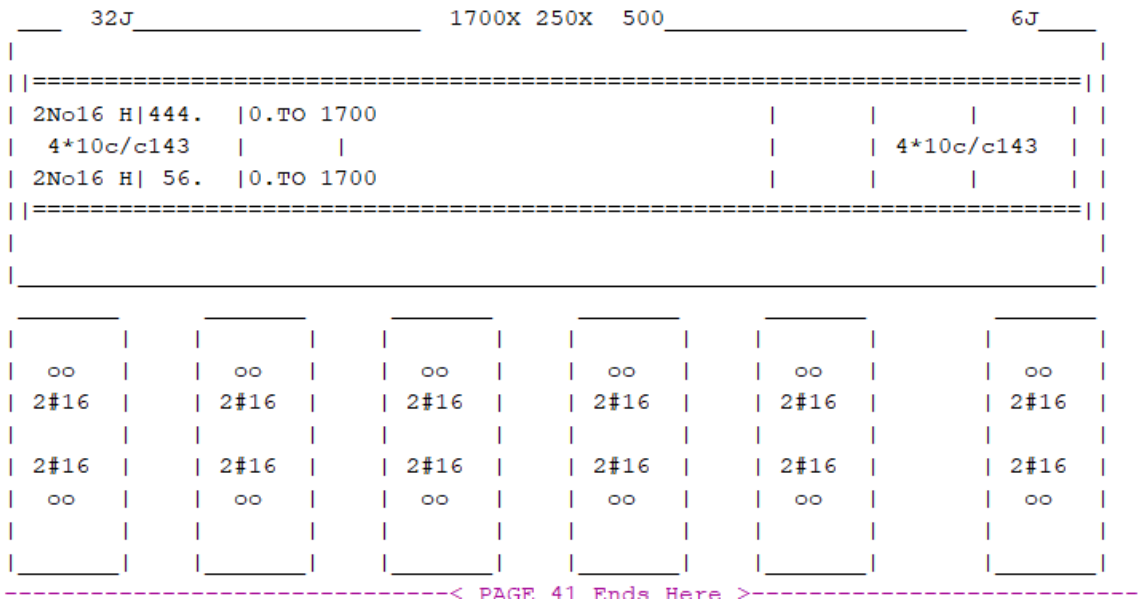
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.2.1. Staad Output for B1

```
            B E A M   N O.    48 D E S I G N   R E S U L T S - SHEAR


   AT START SUPPORT - Vu=   10.05 KNS  Vc=   98.31 KNS  Vs=    0.00 KNS
   Tu=    5.77 KN-MET  Tc=    3.4 KN-MET  Ts=    7.7 KN-MET  LOAD    54
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR  408. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.90 SQ.CM.

   AT END   SUPPORT - Vu=   12.21 KNS  Vc=   95.73 KNS  Vs=    0.00 KNS
   Tu=    5.77 KN-MET  Tc=    3.4 KN-MET  Ts=    7.7 KN-MET  LOAD    54
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR  408. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.90 SQ.CM.
```




```
-------------------------------< PAGE 41 Ends Here >-----------------------------
```

Note: Above result for shear check is for most critical beam in vaporizer Room. Same shear reinforcement detail has also been adopted for Compressor Room.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

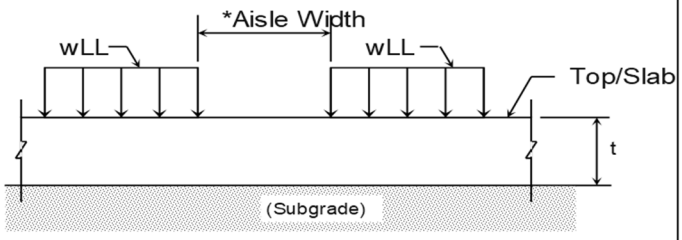
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Allow. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

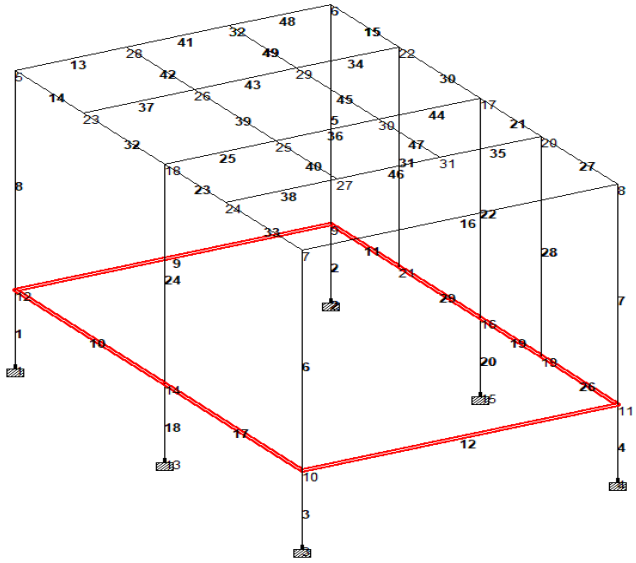
PROPRIETARY & CONFIDENTIAL

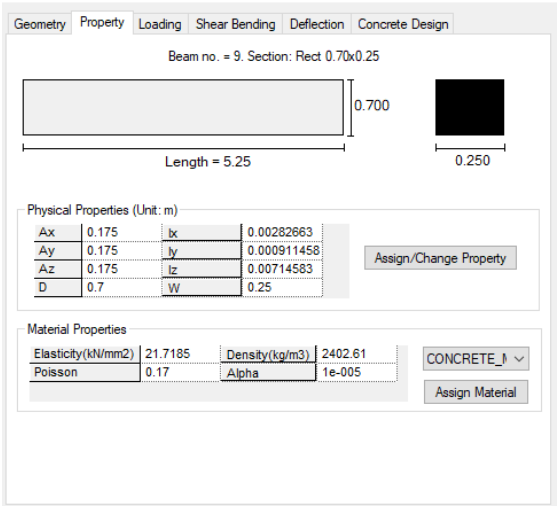
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

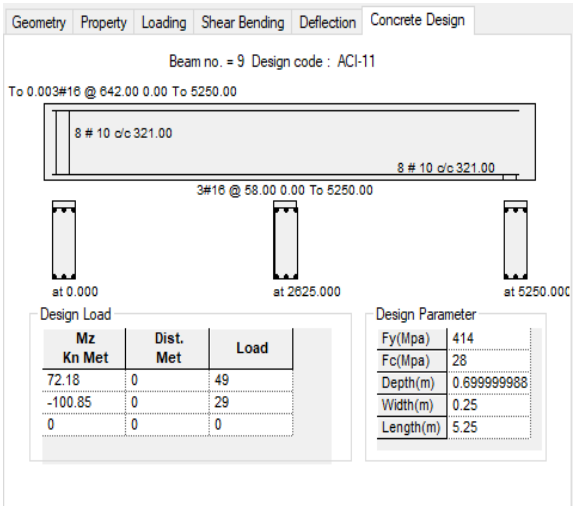
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

11. DESIGN OF GRADE BEAM

11.1. Grade Beam(GB1)





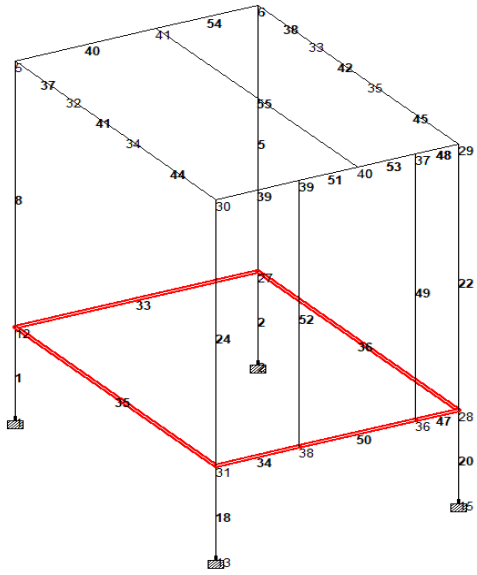


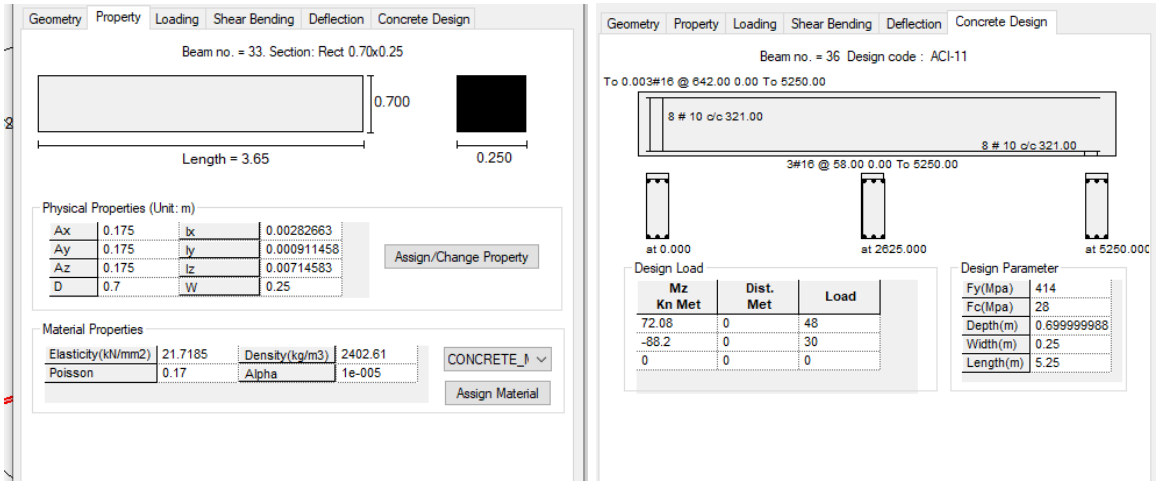
Design of Beam (GB1) _ Vaporizer Room

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.





Design of Beam (GB1) _ Compressor Room

Maximum Moment required from Analysis = 101 **Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=700-50-10-16/2=632mm, Fc'=35 Mpa & ϕ =0.9

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Ast.req = **440 mm²**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **535mm²**

Maximum Moment required from Analysis = **73 Knm (Sagging)**

Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

Ast.pro= **600mm² >(Ast.min, Ast.req)**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

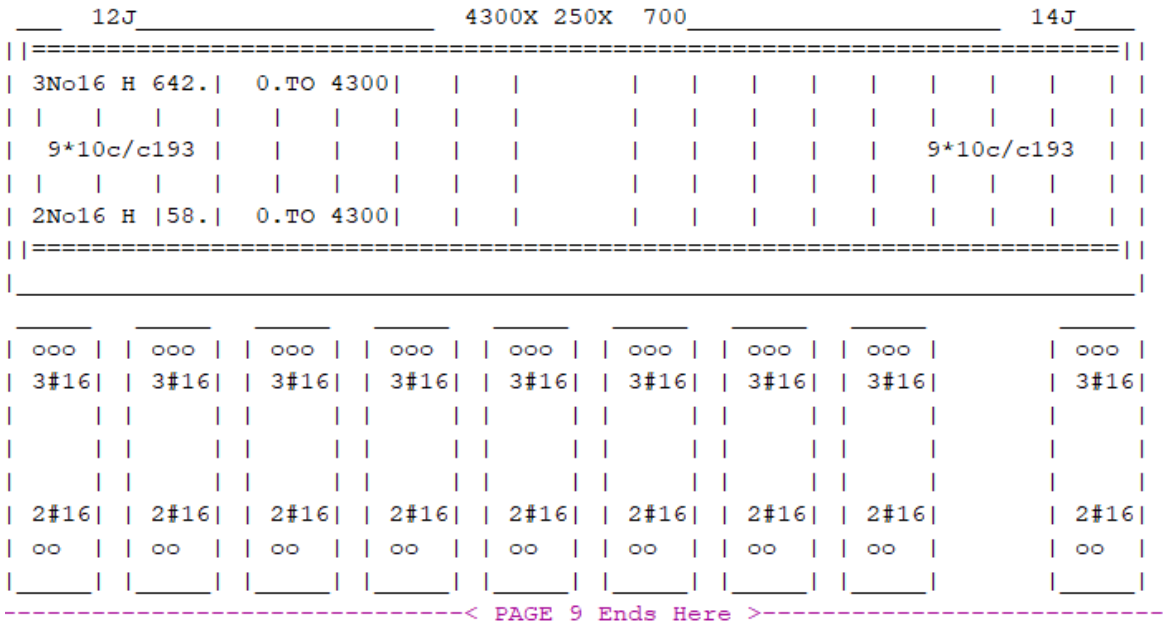
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

11.1.1. Staad Output for GB1

```
                B E A M  N O.    10 D E S I G N  R E S U L T S - SHEAR


   AT START SUPPORT - Vu=    30.16 KNS   Vc=  138.72 KNS   Vs=     0.00 KNS
   Tu=    10.64 KN-MET   Tc=     5.2 KN-MET   Ts=    14.2 KN-MET   LOAD     49
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 193. MM  C/C FOR 1510. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    3.18 SQ.CM.

   AT END    SUPPORT - Vu=    47.12 KNS   Vc=  138.72 KNS   Vs=     0.00 KNS
   Tu=    10.64 KN-MET   Tc=     5.2 KN-MET   Ts=    14.2 KN-MET   LOAD     49
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 193. MM  C/C FOR 1510. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    3.18 SQ.CM.
```




```
-------------------------------< PAGE 9 Ends Here >--------------------------
```

Note: Above result for shear check is for most critical beam in vaporizer Room. Same shear reinforcement detail has also been adopted for Compressor Room.

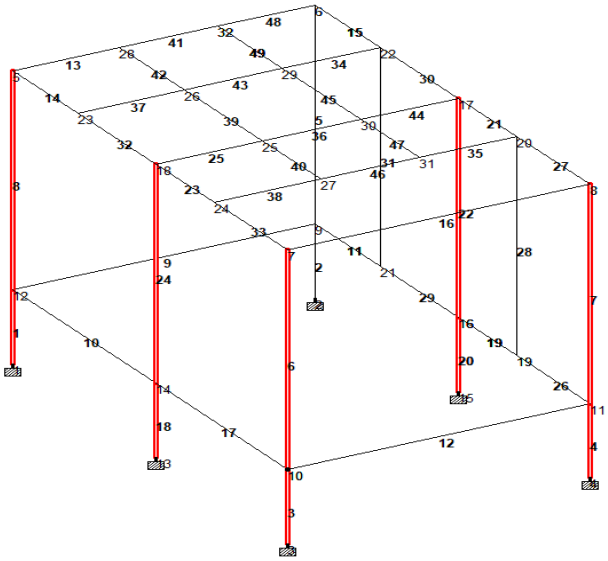
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

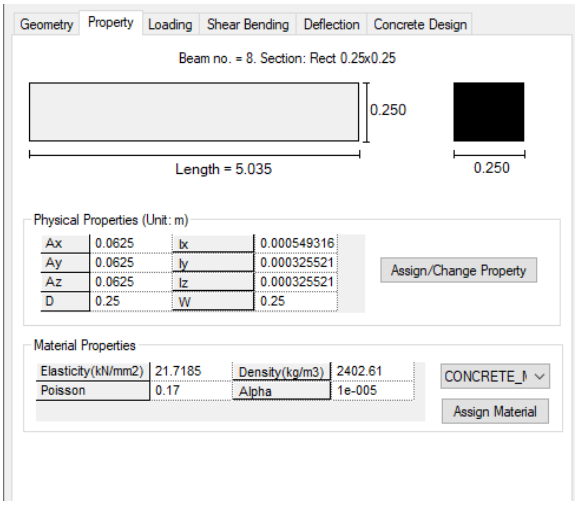
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

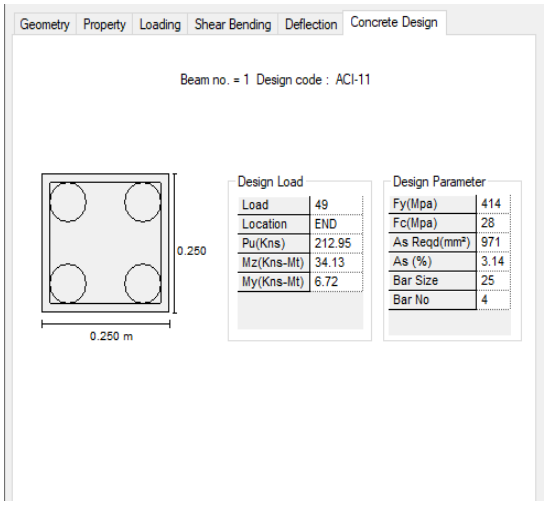
12. CONCRETE COLUMN DESIGN

12.1. Design of Column-(C1)_Vaporizer Room



Column C1





12.1.1. Staad Output for Column C1_Vaporizer Room

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

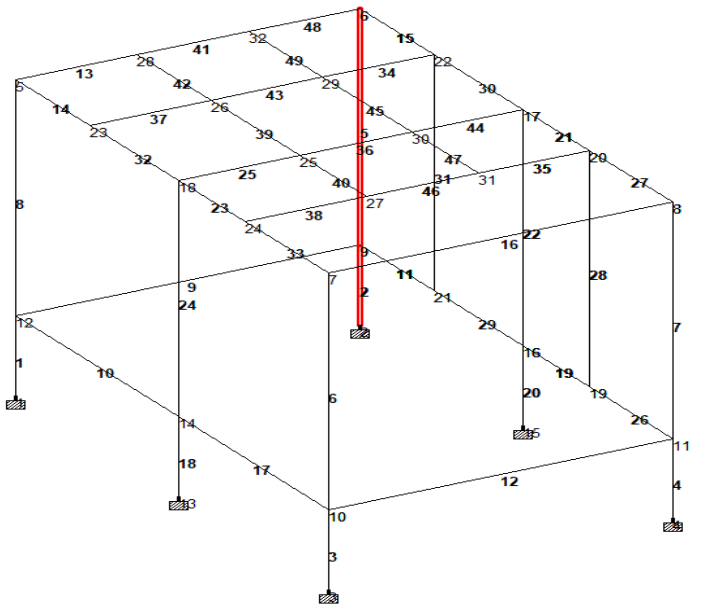
```
        ACI 318-11    COLUMN  NO.      6    DESIGN RESULTS
        ==================================================


    FY - 414.0   FC -   28.0 MPA,   SQRE SIZE - 250.0 X 250.0 MMS, TIED
            AREA OF STEEL REQUIRED =   1521.9   SQ. MM
---------------------------------<  PAGE 44 Ends Here  >----------------------------
     STAAD SPACE                                                 -- PAGE NO.    45


    BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
    ------------------------------------------------------------

      4 - 25 MM                  3.142        49       STA      0.650
    (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
    TIE BAR NUMBER    10 SPACING 250.00 MM
```

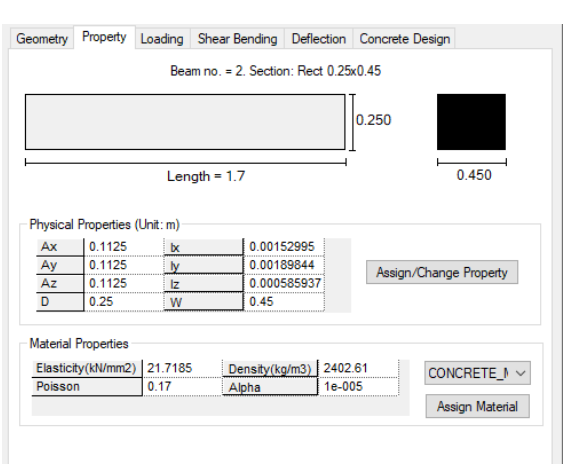
12.2. Design of Column-(C2)_Vaporizer Room

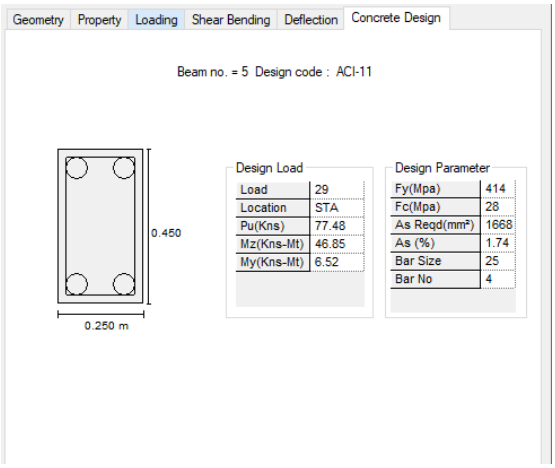


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.





12.2.1. Staad Output for Column C2_Vaporizer Room

```
ACI 318-11   COLUMN  NO.     5   DESIGN RESULTS
==================================================

FY - 414.0  FC -  28.0 MPA,  RECT SIZE - 450.0 X 250.0 MMS, TIED
        AREA OF STEEL REQUIRED =  1668.4  SQ. MM

BAR CONFIGURATION        REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

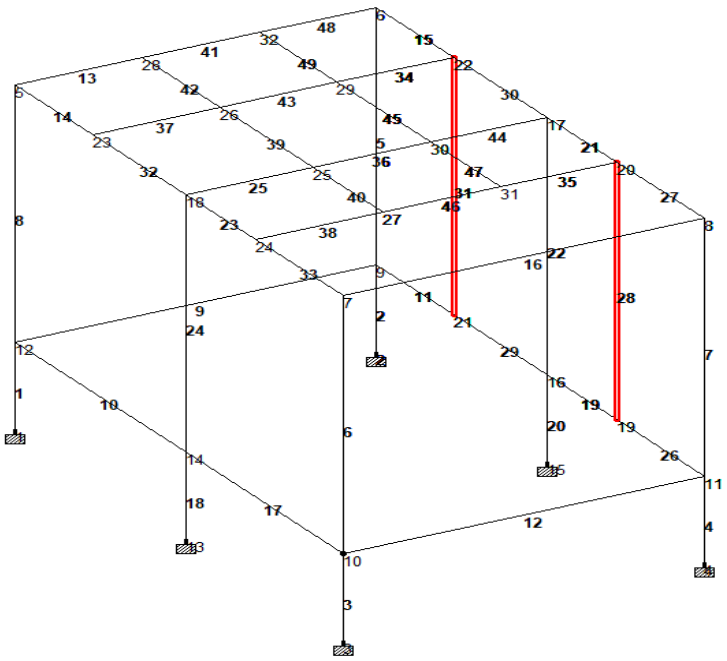
  4 - 25 MM                 1.745       29      STA     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 250.00 MM
```

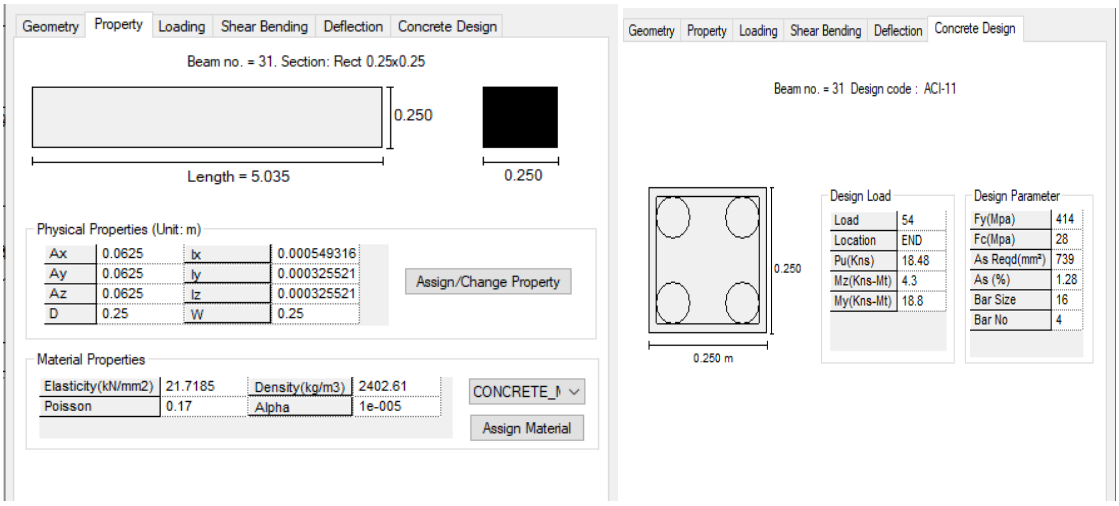

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

12.3. Design of Column-(C3)_Vaporizer Room





PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

12.3.1. Staad Output for Column C3_Vaporizer Room

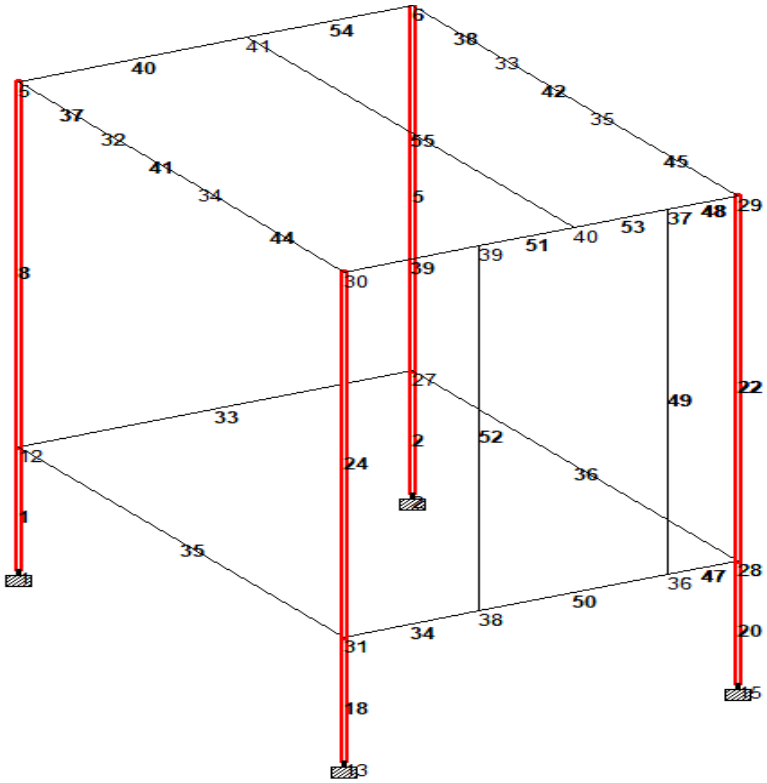
```
ACI 318-11   COLUMN  NO.    31   DESIGN RESULTS
=================================================


FY - 414.0   FC -   28.0 MPA,   SQRE SIZE - 250.0 X 250.0 MMS, TIED
        AREA OF STEEL REQUIRED =    738.8   SQ. MM

BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
-----------------------------------------------------------

  4 - 16 MM                  1.287        54       END      0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    10 SPACING 250.00 MM
```

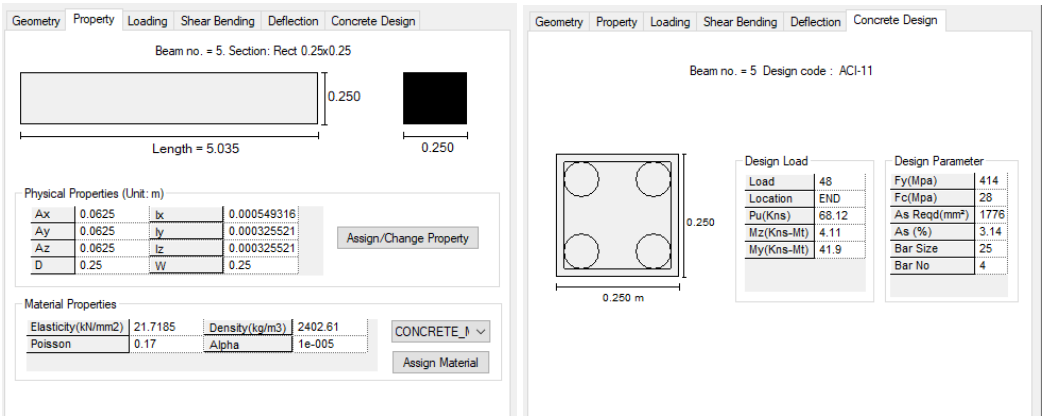
12.4. Design of Column-(C1)_Compressor Room



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



12.4.1. Staad Output for Column C1_Compressor Room

```
                ACI 318-11    COLUMN   NO.      5    DESIGN RESULTS
                ==========================================================


 FY - 414.0   FC -   28.0 MPA,   SQRE SIZE - 250.0 X 250.0 MMS, TIED
          AREA OF STEEL REQUIRED =   1775.6   SQ. MM

 BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
 ------------------------------------------------------------

   4 - 25 MM                  3.142        48       END      0.650
 (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
-------------------------------< PAGE 27 Ends Here >--------------------------
  STAAD SPACE                                                -- PAGE NO.    28

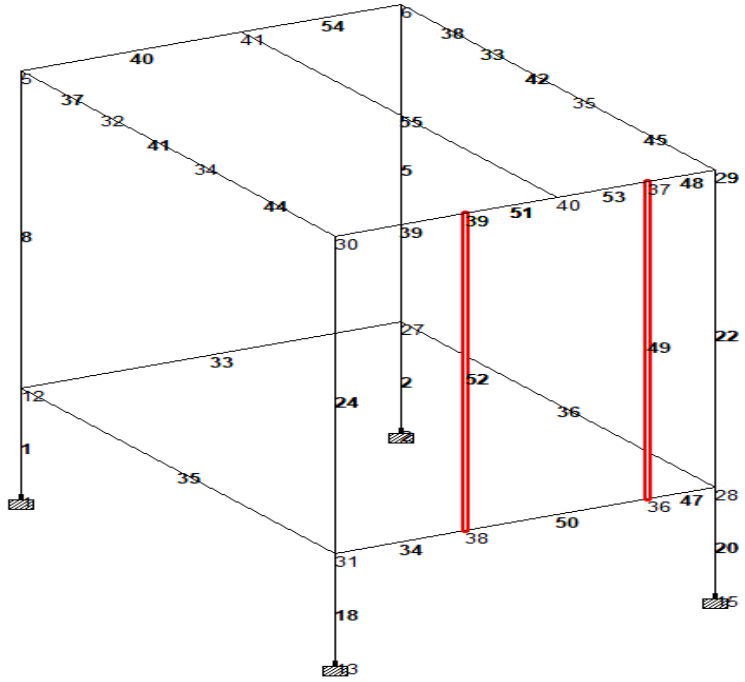
 TIE BAR NUMBER    10 SPACING 250.00 MM
```

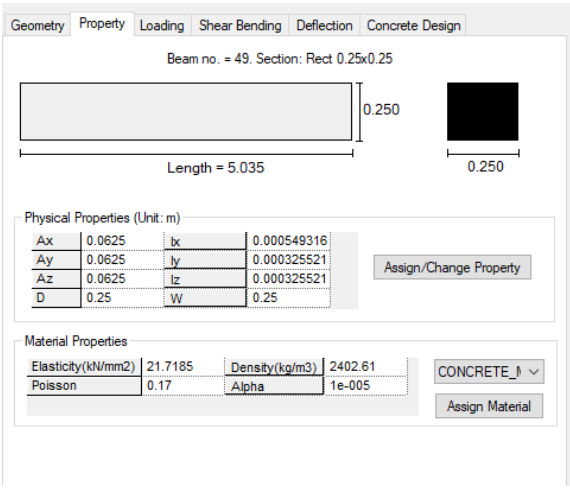

PROPRIETARY & CONFIDENTIAL

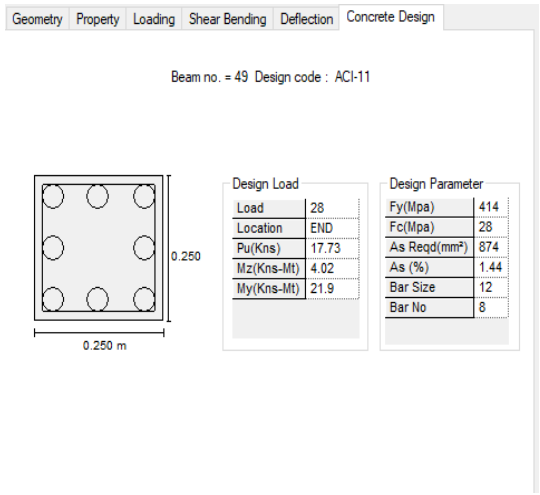
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

12.5. Design of Column-(C2)_Compressor Room







PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

12.5.1. Staad Output for Column C2_Compressor Room

```
        ACI 318-11   COLUMN  NO.    49   DESIGN RESULTS
        ==============================================


 FY - 414.0   FC -   28.0 MPA,   SQRE SIZE - 250.0 X 250.0 MMS, TIED
          AREA OF STEEL REQUIRED =    874.4   SQ. MM
---------------------------------< PAGE 29 Ends Here >----------------------------
  STAAD SPACE                                                  -- PAGE NO.    30


 BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
 ------------------------------------------------------------

   8 - 12 MM                 1.448        28       END      0.650
 (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
 TIE BAR NUMBER    10 SPACING 192.00 MM
```


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

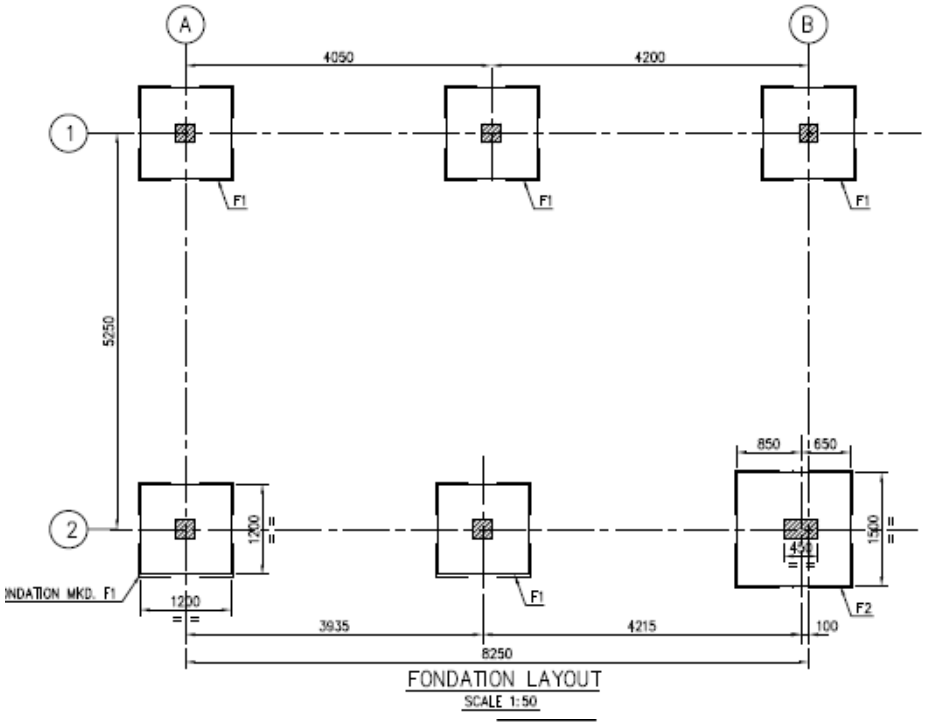
Profile / Borehole	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement
				Settlement	Shear Failure		
-	-	(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)
BH-01	Square Isolated	1.20	1.0 x 1.0	1500	570	570	25
		1.50		2000	850	850	25
		1.20	2.0 x 2.0	1000	750	750	25
		1.50		1200	980	980	25
		1.20	3.0 x 3.0	775	900	775	25
		1.50		950	1120	950	25
		1.20	4.0 x 4.0	625	1040	625	25
		1.50		750	1250	750	25
		1.20	5.0 x 5.0	525	1172	525	25
		1.50		600	1342	600	25

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13.1. Foundation Design Calculation_ Vaporizer Room



FOOTING MARK	LEAN CONCRETE			REINFORCED CONCRETE			BOTTOM REINFORCEMENT		TOP REINFORCEMENT		Ha	REMARKS
	L	W	T	L	W	T	SHORT BARS	LONG BARS	SHORT BARS	LONG BARS		
F1	1400	1400	50	1200	1200	400	ø16@200	ø16@200	–	–	1200	
F2	1700	1700	50	1500	1500	400	ø16@200	ø16@200	–	–	1200	

Foundation Reinforcement Summary

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13.1.1 DESIGN CALCULATION FOR VAPORIZER ROOM FOOTING TYPE F1

	Along-X	Along-Z
Size of column :	0.25m	0.25m
Size of fdn. raft :	1.20m	1.20m
Thickness of raft :	0.40	m
Original ground level:	0.00	m
Finished floor level :	0.50	m
U/S level of foundation :	-1.20	m
U/S level Base Plate :	0.50	m
Unit weight of conc :	24	kN/m^3
Unit weight of soil :	18	kN/m^3

SYMBOLS AND NOTATIONS

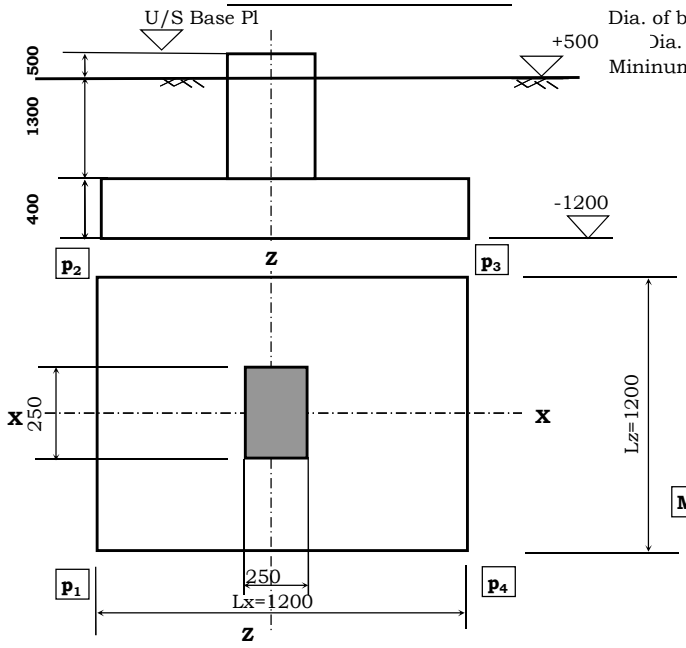
Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

ght of raft & Col.=	(1.2*1.2*0.4*24)+(0.25*0.25*1.3*24) =	15.77KN
Weight of soil =	((1.2*1.2)-(0.25*0.25))*1.3*18 =	32.23KN
	Total =	48.01KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%

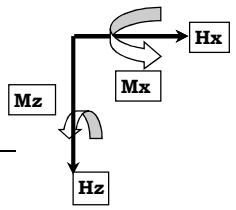


Effective depth of slab along X direction = 342mm = 0.342m

Effective depth of slab along Z direction = 326mm = 0.326m

Clear projection along Z direction = 0.475m

Clear projection along X direction = 0.475m



H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

PROPRIETARY & CONFIDENTIAL

contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part all of the contents in any form without the prior written permission of Outotec is prohibited.

document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either full or in part by such person or body without our prior permission in writing.

13.1.1 DESIGN CALCULATION FOR VAPORIZER ROOM FOOTING TYPE F1

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design										
10 DL	224.0	1.9	-1.5	-1.3	-0.9	271.96	1.87	-1.51	-1.33	-0.95
11 DL+T	224.0	1.9	-1.5	-1.3	-0.9	271.96	1.87	-1.51	-1.33	-0.95
12 DL+LR	230.4	1.8	-1.2	-2.1	-1.6	278.42	1.76	-1.23	-2.13	-1.64
13 D+0.75LR	228.8	1.8	-1.3	-1.9	-1.5	276.80	1.79	-1.30	-1.93	-1.47
14 D+0.7EZ	223.6	1.9	-1.4	-4.8	-4.0	271.58	1.86	-1.44	-4.81	-3.97
15 D+0.7EX	221.3	0.0	2.2	-1.5	-1.1	269.33	-0.01	2.23	-1.53	-1.12
16 D+0.7WX1+0.75LR	224.0	-7.2	11.6	-2.4	-1.9	272.01	-7.22	11.63	-2.41	-1.89
17 D+0.75WZ1+0.75LR	227.8	7.8	-7.3	-13.6	-11.5	275.79	7.81	-7.33	-13.57	-11.54
18 0.6D+WX1	128.0	-10.9	16.3	-1.4	-1.1	175.98	-10.88	16.33	-1.43	-1.13
19 0.6D+WZ1	133.0	9.2	-8.9	-16.3	-14.0	181.03	9.15	-8.95	-16.32	-14.00
20 0.6D+0.7EX	131.7	-0.8	2.8	-1.0	-0.7	179.75	-0.76	2.84	-0.99	-0.74
21 0.6D+0.7EZ	134.0	1.1	-0.8	-4.3	-3.6	182.00	1.11	-0.84	-4.28	-3.59
22 D+WX1	217.6	-10.1	15.7	-2.0	-1.5	265.56	-10.13	15.73	-1.97	-1.51
23 D+WZ1	222.6	9.9	-9.6	-16.9	-14.4	270.61	9.90	-9.55	-16.86	-14.38
39 D+0.75WX2+0.75LR	233.7	9.0	-12.5	-1.4	-1.0	281.71	8.97	-12.47	-1.45	-1.04
40 D+0.75WZ2+0.75LR	229.7	7.9	-7.7	9.9	8.8	277.71	7.89	-7.68	9.95	8.81
41 0.6D+WX2	140.9	10.7	-15.8	-0.2	0.0	188.92	10.70	-15.79	-0.16	0.00
42 0.6D+WZ2	135.6	9.3	-9.4	15.0	13.1	183.59	9.26	-9.40	15.04	13.13
45 D+WX2	230.5	11.5	-16.4	-0.7	-0.4	278.50	11.45	-16.40	-0.69	-0.38
46 D+WZ2	225.2	10.0	-10.0	14.5	12.8	273.17	10.01	-10.01	14.51	12.75

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Strength Design										
24 1.4DL	313.5	2.6	-2.1	-1.9	-1.3	361.54	2.62	-2.12	-1.87	-1.33
25 1.4D+1.4T+0.5LR	314.7	2.7	-2.4	-2.4	-1.8	362.67	2.68	-2.43	-2.36	-1.76
26 1.4D+1.4T+0.5LR	314.7	2.7	-2.4	-2.4	-1.8	362.67	2.68	-2.43	-2.36	-1.76
27 1.2D+0.8WX1+1.6LR	274.0	-7.5	12.4	-3.4	-2.7	321.97	-7.54	12.42	-3.38	-2.70
28 1.2D+1.6WZ1+0.5LR	269.8	15.0	-14.5	-26.8	-23.0	317.82	15.04	-14.54	-26.84	-22.97
29 1.2D+1.6WX1+0.5LR	261.7	-17.0	25.9	-3.0	-2.4	309.74	-17.02	25.91	-3.01	-2.38
30 1.2D+1.6WZ1+0.5LR	269.8	15.0	-14.5	-26.8	-23.0	317.82	15.04	-14.54	-26.84	-22.97
31 1.2D+EX	265.0	-0.4	3.5	-1.9	-1.4	313.00	-0.45	3.54	-1.87	-1.38
32 1.2D+EZ	268.2	2.2	-1.7	-6.6	-5.5	316.21	2.22	-1.72	-6.57	-5.46
33 0.9D+1.6WX1	191.3	-17.5	26.2	-2.2	-1.7	239.33	-17.52	26.22	-2.21	-1.75
34 0.9D+1.6WZ1	199.4	14.5	-14.2	-26.0	-22.3	247.40	14.53	-14.22	-26.04	-22.34
35 0.9D+EZ	201.0	1.7	-1.3	-6.2	-5.2	249.03	1.66	-1.26	-6.17	-5.17
36 0.9D+EX	197.8	-1.0	4.0	-1.5	-1.1	245.81	-1.01	3.99	-1.47	-1.10
47 1.2D+0.8WX2+1.6LR	284.3	9.7	-13.3	-2.4	-1.8	332.32	9.73	-13.27	-2.36	-1.79
48 1.2D+1.6WZ2+0.5LR	273.9	15.2	-15.3	23.3	20.4	321.91	15.21	-15.27	23.35	20.44
49 1.2D+1.6WX2+0.5LR	282.4	17.5	-25.5	-1.0	-0.6	330.44	17.52	-25.49	-0.97	-0.57
53 0.9D+1.6WX2	212.0	17.0	-25.2	-0.2	0.1	260.03	17.01	-25.17	-0.17	0.06
54 0.9D+1.6WZ2	203.5	14.7	-15.0	24.1	21.1	251.50	14.70	-14.96	24.14	21.07

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Check for overturning and sliding:

						Overturning		Sliding	
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
10 DL	272.0	1.9	-1.5	-1.3	-0.9	107.8	172.1	58.1	81.6
11 DL+T	272.0	1.9	-1.5	-1.3	-0.9	107.8	172.1	58.1	81.6
12 DL+LR	278.4	1.8	-1.2	-2.1	-1.6	135.4	101.7	63.3	52.3
13 D+0.75LR	276.8	1.8	-1.3	-1.9	-1.5	127.4	113.1	62.0	57.3
14 D+0.7EZ	271.6	1.9	-1.4	-4.8	-4.0	112.8	41.0	58.6	22.6
15 D+0.7EX	269.3	0.0	2.2	-1.5	-1.1	72.4	144.5	7695.3	70.6
16 D+0.7WX1+0.75LR	272.0	-7.2	11.6	-2.4	-1.9	14.0	86.4	15.1	45.2
17 D+0.75WZ1+0.75LR	275.8	7.8	-7.3	-13.6	-11.5	22.6	14.3	14.1	8.1
18 0.6D+WX1	176.0	-10.9	16.3	-1.4	-1.1	6.5	93.5	6.5	49.1
19 0.6D+WZ1	181.0	9.2	-8.9	-16.3	-14.0	12.1	7.8	7.9	4.4
20 0.6D+0.7EX	179.8	-0.8	2.8	-1.0	-0.7	38.0	145.9	94.2	72.5
21 0.6D+0.7EZ	182.0	1.1	-0.8	-4.3	-3.6	130.3	30.4	65.8	17.0
22 D+WX1	265.6	-10.1	15.7	-2.0	-1.5	10.1	105.7	10.5	54.0
23 D+WZ1	270.6	9.9	-9.6	-16.9	-14.4	17.0	11.3	10.9	6.4
39 D+0.75WX2+0.75LR	281.7	9.0	-12.5	-1.4	-1.0	13.6	162.2	12.6	77.8
40 D+0.75WZ2+0.75LR	277.7	7.9	-7.7	9.9	8.8	21.7	18.9	14.1	11.2
41 0.6D+WX2	188.9	10.7	-15.8	-0.2	0.0	7.2	NA	7.1	484.4
42 0.6D+WZ2	183.6	9.3	-9.4	15.0	13.1	11.7	8.4	7.9	4.9
45 D+WX2	278.5	11.5	-16.4	-0.7	-0.4	10.2	440.9	9.7	161.7
46 D+WZ2	273.2	10.0	-10.0	14.5	12.8	16.4	12.9	10.9	7.5
					Min.=	6.5	7.8	6.5	4.4
						OK	OK	OK	OK

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13.1.1 DESIGN CALCULATION FOR VAPORIZER ROOM FOOTING TYPE F1

Load Combination & Calculation of Base pressure

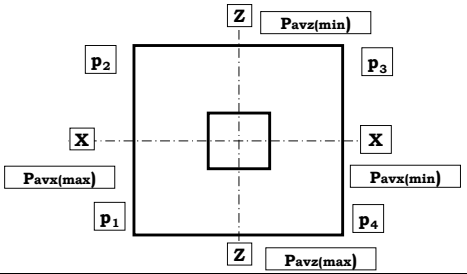
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 350 kN/m^2 For WL/SL case = 437.5 kN/m^2

Gross SBC of soil for DL, LL= 371.6 kN/m^2 For WL/SL case = 459.1 kN/m^2



Allowable Stress Design

Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
	V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
10 DL	271.96	-1.51	-0.95	188.86	5.26	3.29	197.41	190.83	186.90	180.31	175.81	194.12	183.61	192.16	185.57	O.K
11 DL+T	271.96	-1.51	-0.95	188.86	5.26	3.29	197.41	190.83	186.90	180.31	175.81	194.12	183.61	192.16	185.57	O.K
12 DL+LR	278.42	-1.23	-1.64	193.35	4.28	5.70	203.34	191.93	194.77	183.36	181.74	197.64	189.07	199.06	187.65	O.K
13 D+0.75LR	276.80	-1.30	-1.47	192.23	4.53	5.10	201.85	191.65	192.80	182.60	180.25	196.75	187.70	197.33	187.13	O.K
14 D+0.7EZ	271.58	-1.44	-3.97	188.60	5.01	13.78	207.40	179.83	197.37	169.80	185.80	193.62	183.59	202.39	174.82	O.K
15 D+0.7EX	269.33	2.23	-1.12	187.04	7.75	3.88	198.67	190.91	183.17	175.41	177.07	194.79	179.29	190.92	183.16	O.K
16 D+0.7WX1+0.75LR	272.01	11.63	-1.89	188.89	40.37	6.56	235.82	222.70	155.08	141.96	214.22	229.26	148.52	195.45	182.33	O.K
17 D+0.75WZ1+0.75LR	275.79	-7.33	-11.54	191.52	25.46	40.07	257.05	176.91	206.13	125.99	235.45	216.98	166.06	231.59	151.45	O.K
18 0.6D+WX1	175.98	16.33	-1.13	122.21	56.71	3.92	182.84	175.00	69.42	61.58	161.24	178.92	65.50	126.13	118.29	O.K
19 0.6D+WZ1	181.03	-8.95	-14.00	125.71	31.06	48.60	205.38	108.18	143.25	46.05	183.78	156.78	94.65	174.32	77.12	O.K
20 0.6D+0.7EX	179.75	2.84	-0.74	124.83	9.85	2.57	137.25	132.12	117.54	112.41	115.65	134.69	114.98	127.40	122.27	O.K
21 0.6D+0.7EZ	182.00	-0.84	-3.59	126.39	2.91	12.47	141.77	116.83	135.95	111.01	120.17	129.30	123.48	138.86	113.92	O.K
22 D+WX1	265.56	15.73	-1.51	184.42	54.61	5.24	244.26	233.79	135.05	124.57	222.66	239.03	129.81	189.66	179.18	O.K
23 D+WZ1	270.61	-9.55	-14.38	187.92	33.17	49.92	271.01	171.17	204.67	104.84	249.41	221.09	154.76	237.84	138.01	O.K
39 D+0.75WX2+0.75LR	281.71	-12.47	-1.04	195.63	43.28	3.62	242.53	235.29	155.97	148.73	220.93	238.91	152.35	199.25	192.01	O.K
40 D+0.75WZ2+0.75LR	277.71	-7.68	8.81	192.85	26.65	30.58	250.09	188.92	196.78	135.62	228.49	219.51	166.20	223.44	162.27	O.K
41 0.6D+WX2	188.92	-15.79	0.00	131.19	54.83	0.00	186.02	186.02	76.37	76.37	164.42	186.02	76.37	131.20	131.20	O.K
42 0.6D+WZ2	183.59	-9.40	13.13	127.49	32.65	45.60	205.74	114.54	140.44	49.24	184.14	160.14	94.84	173.09	81.89	O.K
45 D+WX2	278.50	-16.40	-0.38	193.40	56.93	1.32	251.65	249.02	137.79	135.16	230.05	250.34	136.48	194.72	192.09	O.K
46 D+WZ2	273.17	-1										224.46	154.95	233.99	145.42	O.K

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Strength Design

24 1.4DL	361.54	-2.12	-1.33	251.07	7.36	4.61	263.04	253.82	248.32	239.10	241.44	258.43	243.71	255.68	246.46	
25 1.4D+1.4T+0.5LR	362.67	-2.43	-1.76	251.85	8.45	6.13	266.43	254.18	249.53	237.28	244.83	260.31	243.41	257.98	245.73	
26 1.4D+1.4T+0.5LR	362.67	-2.43	-1.76	251.85	8.45	6.13	266.43	254.18	249.53	237.28	244.83	260.31	243.41	257.98	245.73	
27 1.2D+0.8WX1+1.6LR	321.97	12.42	-2.70	223.59	43.14	9.36	276.09	257.36	189.81	171.08	254.49	266.73	180.45	232.95	214.22	
28 1.2D+1.6WZ1+0.5LR	317.82	-14.54	-22.97	220.71	50.48	79.76	350.94	191.43	249.99	90.47	329.34	271.19	170.23	300.47	140.95	
29 1.2D+1.6WX1+0.5LR	309.74	25.91	-2.38	215.10	89.96	8.27	313.33	296.79	133.40	116.87	291.73	305.06	125.14	223.37	206.83	
30 1.2D+1.6WZ1+0.5LR	317.82	-14.54	-22.97	220.71	50.48	79.76	350.94	191.43	249.99	90.47	329.34	271.19	170.23	300.47	140.95	
31 1.2D+EX	313.00	3.54	-1.38	217.36	12.27	4.80	234.43	224.84	209.88	200.29	212.83	229.64	205.09	222.16	212.57	
32 1.2D+EZ	316.21	-1.72	-5.46	219.59	5.96	18.94	244.49	206.61	232.58	194.69	222.89	225.55	213.64	238.54	200.65	
33 0.9D+1.6WX1	239.33	26.22	-1.75	166.20	91.06	6.07	263.33	251.18	81.22	69.07	241.73	257.26	75.15	172.28	160.13	
34 0.9D+1.6WZ1	247.40	-14.22	-22.34	171.81	49.39	77.56	298.76	143.63	199.98	44.86	277.16	221.20	122.42	249.37	94.25	
35 0.9D+EZ	249.03	-1.26	-5.17	172.94	4.38	17.95	195.27	159.37	186.51	150.60	173.67	177.32	168.56	190.89	154.99	
36 0.9D+EX	245.81	3.99	-1.10	170.70	13.85	3.81	188.36	180.75	160.66	153.05	166.76	184.56	156.86	174.51	166.90	
47 1.2D+0.8WX2+1.6LR	332.32	-13.27	-1.79	230.78	46.09	6.23	283.09	270.64	190.91	178.46	261.49	276.87	184.69	237.00	224.55	
48 1.2D+1.6WZ2+0.5LR	321.91	-15.27	20.44	223.55	53.02	70.97	347.54	205.60	241.49	99.56	325.94	276.57	170.53	294.52	152.58	
49 1.2D+1.6WX2+0.5LR	330.44	-25.49	-0.57	229.47	88.50	1.99	319.97	315.98	142.97	138.98	298.37	317.98	140.98	231.47	227.48	
53 0.9D+1.6WX2	260.03	-25.17	0.06	180.58	87.41	0.20	268.18	267.78	93.37	92.97	246.58	267.98	93.17	180.78	180.38	
54 0.9D+1.6WZ2	251.50	-14.96	21.07	174.65	51.93	73.16	299.74	153.42	195.88	49.56	278.14	226.58	122.72	247.81	101.49	
Max :	363	26	21													
Min :	176	-25	-23													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 181.74 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 191.74 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 249.41 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 259.41 kN/m^2

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Sample calculation of values of base pressure:

Load case : 11 DL+T

Maximum pressure, p_1 (pmax) = 188.86+5.26+3.29= 197.41 kN/m^2

p_2= 188.86+5.26-3.29= 190.83 kN/m^2

p_3= 188.86-(5.26+3.29)= 180.31 kN/m^2

p_4= 188.86-5.26+3.29= 186.90 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 197.41-1.2*18= 175.81 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (197.41+190.83)/2= 194.12 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (180.31+186.90)/2= 183.60 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (197.41+186.90)/2= 192.15 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (180.31+190.83)/2= 185.57 kN/m^2

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13.1.1 DESIGN CALCULATION FOR VAPORIZER ROOM FOOTING TYPE F1

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.25+2*0.5*0.326)+(0.25+2*0.5*0.326))*2

= 2.304 m

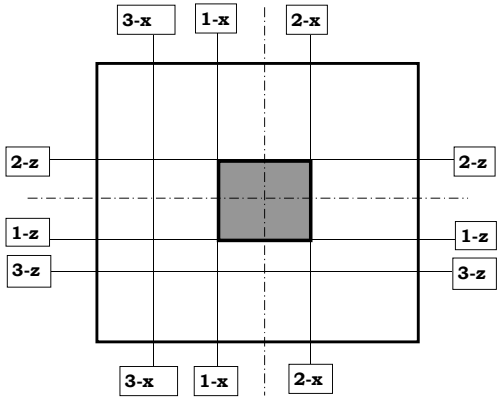
Punching Area= ((0.25+2*0.5*0.326)*(0.25+2*0.5*0.326))

= 0.33 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.3*18)

= 33.00 kN/m^2

Area of footing, A_f= 1.44 m^2



Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions							
24 1.4DL	252.60	249.54	256.80	25.21	23.99	29.87	242.06
25 1.4D+1.4T+0.5LR	253.62	250.10	258.44	25.39	23.99	30.11	242.93
26 1.4D+1.4T+0.5LR	253.62	250.10	258.44	25.39	23.99	30.11	242.93
27 1.2D+0.8WX1+1.6LR	232.58	214.60	257.17	25.08	17.92	30.45	211.55
28 1.2D+1.6WZ1+0.5LR	231.23	210.19	260.00	25.37	16.98	30.94	208.35
29 1.2D+1.6WX1+0.5LR	233.84	196.36	285.12	28.01	13.07	34.86	202.13
30 1.2D+1.6WZ1+0.5LR	231.23	210.19	260.00	25.37	16.98	30.94	208.35
31 1.2D+EX	219.92	214.81	226.92	21.82	19.78	25.97	204.64
32 1.2D+EZ	220.84	218.35	224.23	21.54	20.56	25.52	207.12
33 0.9D+1.6WX1	185.17	147.24	237.08	22.59	7.47	28.48	147.85
34 0.9D+1.6WZ1	182.10	161.52	210.25	19.76	11.56	24.30	154.08
35 0.9D+EZ	173.85	172.03	176.35	16.15	15.42	19.13	155.33
36 0.9D+EX	173.60	167.82	181.49	16.69	14.39	19.95	152.85
47 1.2D+0.8WX2+1.6LR	240.38	221.18	266.65	26.14	18.48	31.76	219.53
48 1.2D+1.6WZ2+0.5LR	234.60	212.50	264.82	25.90	17.09	31.61	211.51
49 1.2D+1.6WX2+0.5LR	247.92	211.04	298.36	29.51	14.82	36.60	218.09
53 0.9D+1.6WX2	198.78	162.37	248.61	23.91	9.39	29.96	163.81
54 0.9D+1.6WZ2	185.47	163.83	215.07	20.29	11.67	24.98	157.23

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Sample calculation of BM & SF

Load case : 24 1.4DL

p_{1x} = 258.43-(258.43-243.71)*0.475/1.2 = 252.60 kN/m^2 Overhang = 0.475 m

p_{2x} = 243.71+(258.43-243.71)*0.475/1.2 = 249.54 kN/m^2 Effective depth = 0.342 m

p_{3x} = 258.43-(258.43-243.71)*(0.475-0.342)/1. 256.80 kN/m^2

BM_{1x} = (252.60-33)*0.475*0.475/2+0.5*(258.43-252.60)*0.475*0.475*0.67 = 25.21 kN-m/m

BM_{2x} = (249.54-33)*0.475*0.475/2-0.5*(-243.71+249.54)*0.475*0.475*0.67 = 23.99 kN-m/m

SF_{3x} = (256.80-33)*(0.475-0.342)+0.5*(258.43-256.80)*(0.475-0.342) = 29.87 kN/m

$S.F_{(punching)}$ = (1.44-0.33)*(251.07-33) = 242.06 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	29.51	0.25	0.34	0.001	275.54	615.60	16ϕT	326.61	300	200	1005.31
At Top	0.00	0.00	0.00	0.000	0.00	615.60	16ϕT	326.61	300	200	1005.31

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 36.60 kN/m

Shear stress , τ_v= (36.6*1000)/(1000*342)

= 0.11 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 242.93 kN

Strength of Concrete in shear Vc1 = 2330.84 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1930.23 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 1553.89 kN **Table 22.6.5.2_ (c)**

Design Vc= 1553.89 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (242.93*1000)/(2304*342)

= 0.31 N/mm^2 **No shear reinforcement reqd.**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions						
24 1.4DL	252.03	250.11	254.54	24.99	24.22	33.09
25 1.4D+1.4T+0.5LR	253.13	250.58	256.46	25.20	24.18	33.41
26 1.4D+1.4T+0.5LR	253.13	250.58	256.46	25.20	24.18	33.41
27 1.2D+0.8WX1+1.6LR	225.54	221.63	230.62	22.28	20.72	29.62
28 1.2D+1.6WZ1+0.5LR	237.33	204.09	280.66	27.82	14.53	38.38
29 1.2D+1.6WX1+0.5LR	216.82	213.38	221.32	21.23	19.85	28.21
30 1.2D+1.6WZ1+0.5LR	237.33	204.09	280.66	27.82	14.53	38.38
31 1.2D+EX	218.36	216.37	220.97	21.20	20.40	28.10
32 1.2D+EZ	223.54	215.65	233.84	22.63	19.47	30.27
33 0.9D+1.6WX1	167.47	164.94	170.77	15.53	14.52	20.64
34 0.9D+1.6WZ1	187.97	155.65	230.11	22.12	9.20	30.80
35 0.9D+EZ	176.68	169.20	186.43	17.28	14.29	23.19
36 0.9D+EX	171.50	169.91	173.57	15.85	15.22	21.01
47 1.2D+0.8WX2+1.6LR	232.07	229.48	235.45	22.83	21.79	30.28
48 1.2D+1.6WZ2+0.5LR	238.34	208.76	276.90	27.41	15.58	37.65
49 1.2D+1.6WX2+0.5LR	229.89	229.06	230.97	22.33	22.00	29.54
53 0.9D+1.6WX2	180.62	180.54	180.73	16.67	16.63	22.02
54 0.9D+1.6WZ2	189.89	159.41	229.64	22.08	9.88	30.65

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Sample calculation of BM & SF

Load case : 24 1.4DL

p_{1z} = 255.68-(255.68-246.46)*0.475/1.2= 252.03 kN/m^2

p_{2z} = 246.46+(255.68-246.46)*0.475/1.2= 250.11 kN/m^2

p_{3z} = 255.68-(255.68-246.46)*(0.475-0.326)/1. 254.54 kN/m^2

Overhang = 0.475 m

Effective depth = 0.326 m

BM_{1z} = (252.03-33)*0.475*0.475/2+0.5*(255.68-252.03)*0.475*0.475*0.67 = 24.99 kN-m/m

BM_{2z} = (250.11-33)*0.475*0.475/2-0.5*(-246.46+250.11)*0.475*0.475*0.67 = 24.22 kN-m/m

SF_{3z} = (254.54-33)*(0.475-0.326)+0.5*(255.68-254.54)*(0.475-0.326) = 33.09 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	27.82	0.262	0.349	0.0008	285.93	586.80	16ϕT	342.64	300	200	1005.3
Top	0.00	0.000	0.000	0.0000	0.00	586.80	16ϕT	342.64	300	200	1005.3

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 38.38 kN/m

Shear stress , τ_v= (38.38*1000)/(1000*326)

= 0.12 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **16ϕT** **200** c/c

At Top **16ϕT** **200** c/c

Along Z Direction :

At Bottom **16ϕT** **200** c/c

At Top **16ϕT** **200** c/c

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13.1.2 DESIGN CALCULATION FOR VAPORIZER ROOM FOOTING TYPE F2

	Along-X	Along-Z
Size of column :	0.25m	0.45m
Size of fdn. raft :	1.50m	1.50m
Thickness of raft :	0.40	m
Original ground level:	0.00	m
Finished floor level :	0.50	m
U/S level of foundation :	-1.20	m
U/S level Base Plate :	0.50	m
Unit weight of conc :	24	kN/m^3
Unit weight of soil :	18	kN/m^3

SYMBOLS AND NOTATIONS

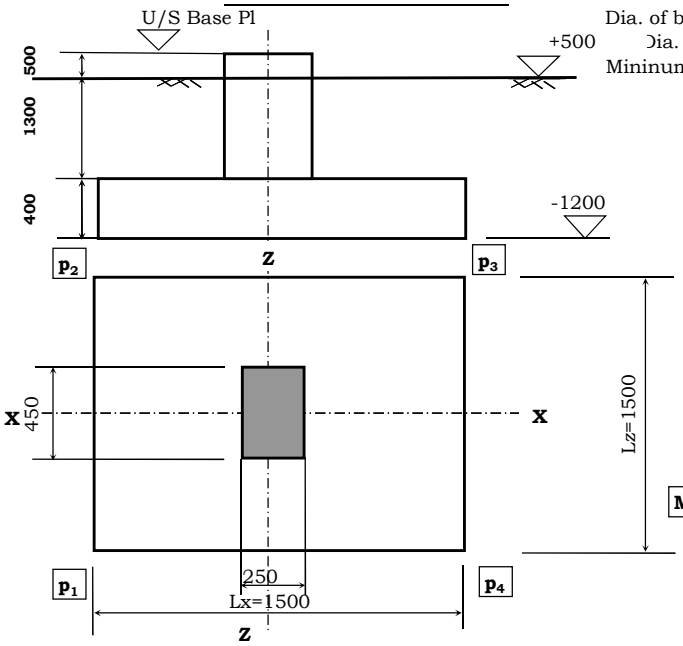
Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

ght of raft & Col.=	(1.5*1.5*0.4*24)+(0.25*0.45*1.3*24) =	25.11KN
Weight of soil =	((1.5*1.5)-(0.25*0.45))*1.3*18 =	50.02KN
	Total =	75.13KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%

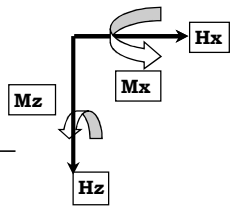


Effective depth of slab along X direction = 342mm = 0.342m

Effective depth of slab along Z direction = 326mm = 0.326m

Clear projection along Z direction = 0.525m

Clear projection along X direction = 0.625m



H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

PROPRIETARY & CONFIDENTIAL

contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part all of the contents in any form without the prior written permission of Outotec is prohibited.

document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either full or in part by such person or body without our prior permission in writing.

13.1.2 DESIGN CALCULATION FOR VAPORIZER ROOM FOOTING TYPE F2

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design										
10 DL	144.0	-14.3	6.5	9.4	3.3	219.09	-14.33	6.54	9.35	3.35
11 DL+T	144.0	-14.3	6.5	9.4	3.3	219.09	-14.33	6.54	9.35	3.35
12 DL+LR	159.0	-15.9	8.5	6.1	-7.7	234.09	-15.94	8.47	6.10	-7.73
13 D+0.75LR	155.2	-15.5	8.0	6.9	-5.7	230.34	-15.54	7.99	6.91	-5.68
14 D+0.7EZ	134.8	-13.7	5.8	3.6	-8.5	209.90	-13.74	5.84	3.61	-8.45
15 D+0.7EX	151.9	-19.0	12.0	9.6	4.2	227.00	-18.98	11.96	9.63	4.17
16 D+0.7WX1+0.75LR	171.5	-27.5	21.2	7.7	-5.9	246.59	-27.50	21.20	7.66	-5.88
17 D+0.75WZ1+0.75LR	133.9	-14.0	6.3	-13.8	-33.3	209.00	-14.01	6.32	-13.79	-33.31
18 0.6D+WX1	108.0	-24.5	21.5	6.6	4.9	183.18	-24.55	21.53	6.61	4.93
19 0.6D+WZ1	57.9	-6.6	1.7	-22.0	-36.6	133.05	-6.56	1.70	-21.98	-36.56
20 0.6D+0.7EX	94.3	-13.3	9.3	5.9	2.8	169.41	-13.25	9.34	5.89	2.83
21 0.6D+0.7EZ	77.2	-8.0	3.2	-0.1	-8.6	152.31	-8.01	3.23	-0.13	-8.64
22 D+WX1	165.6	-30.3	24.1	10.4	6.3	240.76	-30.28	24.15	10.35	6.27
23 D+WZ1	115.5	-12.3	4.3	-18.2	-36.4	190.64	-12.29	4.31	-18.24	-36.36
39 D+0.75WX2+0.75LR	138.6	-3.4	-5.4	6.4	-5.3	213.69	-3.45	-5.43	6.38	-5.25
40 D+0.75WZ2+0.75LR	177.8	-16.9	9.7	27.8	26.0	252.95	-16.87	9.67	27.82	25.98
41 0.6D+WX2	64.2	7.5	-14.0	4.9	-0.6	139.31	7.53	-13.97	4.90	-0.65
42 0.6D+WZ2	116.5	-10.4	6.2	33.5	40.1	191.65	-10.37	6.17	33.49	40.08
45 D+WX2	121.8	1.8	-11.4	8.6	0.7	196.90	1.79	-11.36	8.65	0.69
46 D+WZ2	174.1	-16.1	8.8	37.2	41.4	249.24	-16.10	8.78	37.23	41.42

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Strength Design										
24 1.4DL	201.5	-20.1	9.1	13.1	4.7	276.68	-20.06	9.15	13.10	4.69
25 1.4D+1.4T+0.5LR	211.3	-20.5	10.0	10.5	-3.0	286.45	-20.49	10.02	10.50	-3.05
26 1.4D+1.4T+0.5LR	211.3	-20.5	10.0	10.5	-3.0	286.45	-20.49	10.02	10.50	-3.05
27 1.2D+0.8WX1+1.6LR	214.1	-32.5	25.0	6.8	-12.8	289.21	-32.53	25.03	6.81	-12.78
28 1.2D+1.6WZ1+0.5LR	134.7	-14.7	5.3	-34.6	-62.5	209.85	-14.74	5.25	-34.56	-62.48
29 1.2D+1.6WX1+0.5LR	214.9	-43.5	37.0	11.2	4.7	290.05	-43.52	36.99	11.19	4.74
30 1.2D+1.6WZ1+0.5LR	134.7	-14.7	5.3	-34.6	-62.5	209.85	-14.74	5.25	-34.56	-62.48
31 1.2D+EX	184.1	-23.8	15.6	11.6	5.2	259.18	-23.84	15.58	11.62	5.19
32 1.2D+EZ	159.6	-16.4	6.9	3.0	-12.2	234.75	-16.36	6.85	3.02	-12.18
33 0.9D+1.6WX1	164.2	-38.4	34.1	10.0	7.7	239.37	-38.41	34.06	10.02	7.68
34 0.9D+1.6WZ1	84.0	-9.6	2.3	-35.7	-58.5	159.17	-9.64	2.32	-35.74	-58.52
35 0.9D+EZ	116.4	-12.1	4.9	0.2	-12.3	191.56	-12.06	4.89	0.22	-12.33
36 0.9D+EX	140.9	-19.5	13.6	8.8	4.2	215.99	-19.55	13.62	8.82	4.18
47 1.2D+0.8WX2+1.6LR	179.0	-6.9	-3.4	5.4	-12.1	254.12	-6.88	-3.37	5.45	-12.11
48 1.2D+1.6WZ2+0.5LR	228.5	-20.8	12.4	54.2	61.0	303.62	-20.84	12.40	54.20	60.98
49 1.2D+1.6WX2+0.5LR	144.7	7.8	-19.8	8.5	-2.6	219.87	7.80	-19.81	8.46	-2.62
53 0.9D+1.6WX2	94.1	12.9	-22.7	7.3	-1.2	169.19	12.90	-22.74	7.28	-1.23
54 0.9D+1.6WZ2	177.8	-15.7	9.5	53.0	63.9	252.93	-15.73	9.48	53.03	63.92

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Check for overturning and sliding:

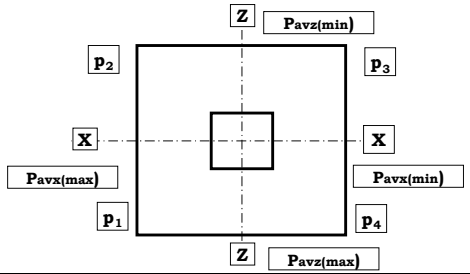
						Overturning		Sliding	
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
10 DL	219.1	-14.3	6.5	9.4	3.3	25.1	49.1	6.1	9.4
11 DL+T	219.1	-14.3	6.5	9.4	3.3	25.1	49.1	6.1	9.4
12 DL+LR	234.1	-15.9	8.5	6.1	-7.7	20.7	22.7	5.9	15.4
13 D+0.75LR	230.3	-15.5	8.0	6.9	-5.7	21.6	30.4	5.9	13.3
14 D+0.7EZ	209.9	-13.7	5.8	3.6	-8.5	26.9	18.6	6.1	23.2
15 D+0.7EX	227.0	-19.0	12.0	9.6	4.2	14.2	40.9	4.8	9.4
16 D+0.7WX1+0.75LR	246.6	-27.5	21.2	7.7	-5.9	8.7	31.4	3.6	12.9
17 D+0.75WZ1+0.75LR	209.0	-14.0	6.3	-13.8	-33.3	24.8	4.7	6.0	6.1
18 0.6D+WX1	183.2	-24.5	21.5	6.6	4.9	6.4	27.9	3.0	11.1
19 0.6D+WZ1	133.1	-6.6	1.7	-22.0	-36.6	58.8	2.7	8.1	2.4
20 0.6D+0.7EX	169.4	-13.3	9.3	5.9	2.8	13.6	44.9	5.1	11.5
21 0.6D+0.7EZ	152.3	-8.0	3.2	-0.1	-8.6	35.4	13.2	7.6	472.3
22 D+WX1	240.8	-30.3	24.1	10.4	6.3	7.5	28.8	3.2	9.3
23 D+WZ1	190.6	-12.3	4.3	-18.2	-36.4	33.2	3.9	6.2	4.2
39 D+0.75WX2+0.75LR	213.7	-3.4	-5.4	6.4	-5.3	29.5	30.5	24.8	13.4
40 D+0.75WZ2+0.75LR	252.9	-16.9	9.7	27.8	26.0	19.6	7.3	6.0	3.6
41 0.6D+WX2	139.3	7.5	-14.0	4.9	-0.6	7.5	161.8	7.4	11.4
42 0.6D+WZ2	191.7	-10.4	6.2	33.5	40.1	23.3	3.6	7.4	2.3
45 D+WX2	196.9	1.8	-11.4	8.6	0.7	13.0	212.6	43.9	9.1
46 D+WZ2	249.2	-16.1	8.8	37.2	41.4	21.3	4.5	6.2	2.7
					Min.=	6.4	2.7	3.0	2.3
						OK	OK	OK	OK

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13.1.2 DESIGN CALCULATION FOR VAPORIZER ROOM FOOTING TYPE F2



Load Combination & Calculation of Base pressure

Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 350 kN/m^2 For WL/SL case = 437.5 kN/m^2

Gross SBC of soil for DL, LL= 371.6 kN/m^2 For WL/SL case = 459.1 kN/m^2

Allowable Stress Design

Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
	V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
10 DL	219.09	6.54	3.35	97.37	11.62	5.95	114.94	103.04	91.71	79.80	93.34	108.99	85.76	103.33	91.42	O.K
11 DL+T	219.09	6.54	3.35	97.37	11.62	5.95	114.94	103.04	91.71	79.80	93.34	108.99	85.76	103.33	91.42	O.K
12 DL+LR	234.09	8.47	-7.73	104.04	15.06	13.75	132.85	105.35	102.72	75.23	111.25	119.10	88.98	117.79	90.29	O.K
13 D+0.75LR	230.34	7.99	-5.68	102.37	14.20	10.09	126.67	106.48	98.27	78.08	105.07	116.58	88.18	112.47	92.28	O.K
14 D+0.7EZ	209.90	5.84	-8.45	93.29	10.39	15.02	118.70	88.65	97.92	67.88	97.10	103.68	82.90	108.31	78.27	O.K
15 D+0.7EX	227.00	11.96	4.17	100.89	21.25	7.41	129.55	114.73	87.04	72.23	107.95	122.14	79.64	108.30	93.48	O.K
16 D+0.7WX1+0.75LR	246.59	21.20	-5.88	109.60	37.68	10.46	157.74	136.82	82.37	61.46	136.14	147.28	71.92	120.06	99.14	O.K
17 D+0.75WZ1+0.75LR	209.00	6.32	-33.31	92.89	11.24	59.22	163.34	44.90	140.87	22.43	141.74	104.12	81.65	152.11	33.67	O.K
18 0.6D+WX1	183.18	21.53	4.93	81.41	38.28	8.76	128.45	110.93	51.89	34.37	106.85	119.69	43.13	90.17	72.65	O.K
19 0.6D+WZ1	133.05	1.70	-36.56	59.13	3.02	64.99	127.14	0.00	125.00	0.00	110.00	63.57	62.50	126.07	0.00	O.K
20 0.6D+0.7EX	169.41	9.34	2.83	75.30	16.61	5.03	96.93	86.88	63.72	53.66	75.33	91.91	58.69	80.33	70.27	O.K
21 0.6D+0.7EZ	152.31	3.23	-8.64	67.69	5.74	15.37	88.80	58.07	77.32	46.59	67.20	73.44	61.96	83.06	52.33	O.K
22 D+WX1	240.76	24.15	6.27	107.00	42.93	11.14	161.07	138.79	75.22	52.94	139.47	149.93	64.08	118.15	95.87	O.K
23 D+WZ1	190.64	4.31	-36.36	84.73	7.66	64.64	157.04	27.75	141.71	12.42	135.44	92.40	77.07	149.38	20.09	O.K
39 D+0.75WX2+0.75LR	213.69	-5.43	-5.25	94.97	9.65	9.33	113.96	95.29	94.66	75.99	92.36	104.63	85.33	104.31	85.64	O.K
40 D+0.75WZ2+0.75LR	252.95	9.67	25.98	112.42	17.19	46.19	175.80	83.43	141.41	49.04	154.20	129.62	95.23	158.61	66.24	O.K
41 0.6D+WX2	139.31	-13.97	-0.65	61.92	24.83	1.15	87.90	85.60	38.23	35.94	66.30	86.75	37.09	63.07	60.77	O.K
42 0.6D+WZ2	191.65	6.17	40.08	85.18	10.96	71.25	167.39	24.90	145.46	2.97	145.79	96.15	74.22	156.43	13.94	O.K
45 D+WX2	196.90	-11.36	0.69	87.51	20.19	1.23	108.93	106.46	68.56	66.09	87.33	107.70	67.33	88.75	86.28	O.K
46 D+WZ2	249.24											126.38	95.16	184.40	37.14	O.K

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Strength Design

24 1.4DL	276.68	9.15	4.69	122.97	16.26	8.33	147.56	130.90	115.04	98.37	125.96	139.23	106.71	131.30	114.64	O.K
25 1.4D+1.4T+0.5LR	286.45	10.02	-3.05	127.31	17.82	5.42	150.55	139.71	114.91	104.08	128.95	145.13	109.50	132.73	121.90	O.K
26 1.4D+1.4T+0.5LR	286.45	10.02	-3.05	127.31	17.82	5.42	150.55	139.71	114.91	104.08	128.95	145.13	109.50	132.73	121.90	O.K
27 1.2D+0.8WX1+1.6LR	289.21	25.03	-12.78	128.54	44.50	22.72	195.76	150.31	106.77	61.32	174.16	173.04	84.05	151.27	105.82	O.K
28 1.2D+1.6WZ1+0.5LR	209.85	5.25	-62.48	93.27	9.34	111.08	220.00	0.00	210.00	0.00	198.40	110.00	105.00	215.00	0.00	O.K
29 1.2D+1.6WX1+0.5LR	290.05	36.99	4.74	128.91	65.76	8.43	203.10	186.24	71.58	54.73	181.50	194.67	63.16	137.34	120.49	O.K
30 1.2D+1.6WZ1+0.5LR	209.85	5.25	-62.48	93.27	9.34	111.08	220.00	0.00	210.00	0.00	198.40	110.00	105.00	215.00	0.00	O.K
31 1.2D+EX	259.18	15.58	5.19	115.19	27.70	9.22	152.12	133.67	96.71	78.26	130.52	142.90	87.49	124.42	105.97	O.K
32 1.2D+EZ	234.75	6.85	-12.18	104.33	12.18	21.66	138.18	94.86	113.81	70.49	116.58	116.52	92.15	126.00	82.68	O.K
33 0.9D+1.6WX1	239.37	34.06	7.68	106.38	60.55	13.65	180.59	153.28	59.49	32.18	158.99	166.94	45.84	120.04	92.73	O.K
34 0.9D+1.6WZ1	159.17	2.32	-58.52	70.74	4.13	104.03	200.00	0.00	180.00	0.00	178.40	100.00	90.00	190.00	0.00	O.K
35 0.9D+EZ	191.56	4.89	-12.33	85.14	8.70	21.91	115.75	71.92	98.35	54.52	94.15	93.84	76.44	107.05	63.22	O.K
36 0.9D+EX	215.99	13.62	4.18	96.00	24.22	7.44	127.65	112.78	79.21	64.34	106.05	120.22	71.78	103.43	88.56	O.K
47 1.2D+0.8WX2+1.6LR	254.12	-3.37	-12.11	112.94	5.99	21.52	140.46	97.41	128.47	85.43	118.86	118.94	106.95	134.47	91.42	O.K
48 1.2D+1.6WZ2+0.5LR	303.62	12.40	60.98	134.94	22.05	108.41	265.40	48.58	221.30	4.48	243.80	156.99	112.89	243.35	26.53	O.K
49 1.2D+1.6WX2+0.5LR	219.87	-19.81	-2.62	97.72	35.22	4.65	137.60	128.30	67.15	57.85	116.00	132.95	62.50	102.38	93.08	O.K
53 0.9D+1.6WX2	169.19	-22.74	-1.23	75.19	40.43	2.19	117.82	113.43	36.95	32.57	96.22	115.63	34.76	77.39	73.00	O.K
54 0.9D+1.6WZ2	252.93	9.48	63.92	112.41	16.84	113.64	242.90	15.62	209.21	0.00	221.30	129.26	104.61	226.06	7.81	O.K
Max :	304	37	64													
Min :	133	-23	-62													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 111.25 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 121.25 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 178.41 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 188.41 kN/m^2

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Sample calculation of values of base pressure:

Load case : 11 DL+T

Maximum pressure, p_1 (pmax) = 97.37+11.62+5.95= 114.94 kN/m^2

p_2= 97.37+11.62-5.95= 103.04 kN/m^2

p_3= 97.37-(11.62+5.95)= 79.80 kN/m^2

p_4= 97.37-11.62+5.95= 91.71 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 114.94-1.2*18= 93.34 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (114.94+103.04)/2= 108.99 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (79.80+91.71)/2= 85.76 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (114.94+91.71)/2= 103.33 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (79.80+103.04)/2= 91.42 kN/m^2

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13.1.2 DESIGN CALCULATION FOR VAPORIZER ROOM FOOTING TYPE F2

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.25+2*0.5*0.326)+(0.45+2*0.5*0.326))*2

= 2.704 m

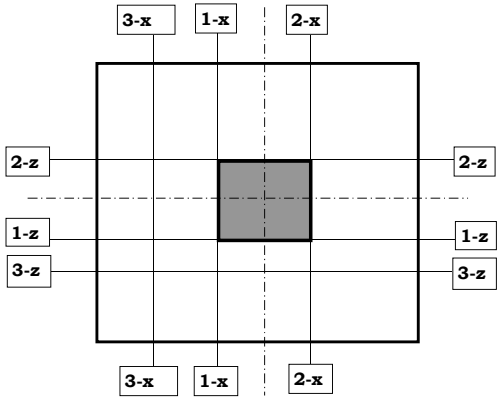
Punching Area= ((0.25+2*0.5*0.326)*(0.45+2*0.5*0.326))

= 0.45 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.3*18)

= 33.00 kN/m^2

Area of footing, A_f= 2.25 m^2



Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions							
24 1.4DL	125.68	120.26	133.09	19.87	15.28	29.19	161.94
25 1.4D+1.4T+0.5LR	130.28	124.35	138.41	20.93	15.91	30.78	169.76
26 1.4D+1.4T+0.5LR	130.28	124.35	138.41	20.93	15.91	30.78	169.76
27 1.2D+0.8WX1+1.6LR	135.96	121.13	156.25	24.94	12.38	37.26	171.97
28 1.2D+1.6WZ1+0.5LR	107.92	107.08	109.06	14.90	14.20	21.66	108.48
29 1.2D+1.6WX1+0.5LR	139.87	117.96	169.86	28.01	9.46	42.24	172.64
30 1.2D+1.6WZ1+0.5LR	107.92	107.08	109.06	14.90	14.20	21.66	108.48
31 1.2D+EX	119.81	110.58	132.45	19.96	12.15	29.62	147.94
32 1.2D+EZ	106.37	102.30	111.92	15.65	12.21	22.99	128.40
33 0.9D+1.6WX1	116.48	96.30	144.09	22.88	5.79	34.67	132.09
34 0.9D+1.6WZ1	95.83	94.17	98.11	12.81	11.40	18.69	67.93
35 0.9D+EZ	86.59	83.69	90.56	11.41	8.96	16.75	93.85
36 0.9D+EX	100.04	91.96	111.08	15.72	8.89	23.39	113.39
47 1.2D+0.8WX2+1.6LR	113.94	111.95	116.68	16.46	14.77	24.00	143.90
48 1.2D+1.6WZ2+0.5LR	138.62	131.27	148.67	23.02	16.80	33.91	183.49
49 1.2D+1.6WX2+0.5LR	103.60	91.85	119.66	17.61	7.67	26.41	116.50
53 0.9D+1.6WX2	81.93	68.46	100.37	13.94	2.54	21.23	75.95
54 0.9D+1.6WZ2	118.99	114.88	124.61	18.13	14.66	26.58	142.94

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Sample calculation of BM & SF

Load case : 24 1.4DL

p_{1x} = 139.23-(139.23-106.71)*0.625/1.5 = 125.68 kN/m² Overhang = 0.625 m

p_{2x} = 106.71+(139.23-106.71)*0.625/1.5 = 120.26 kN/m² Effective depth = 0.342 m

p_{3x} = 139.23-(139.23-106.71)*(0.625-0.342)/1. 133.09 kN/m²

BM_{1x} = (125.68-33)*0.625*0.625/2+0.5*(139.23-125.68)*0.625*0.625*0.67 = 19.87 kN-m/m

BM_{2x} = (120.26-33)*0.625*0.625/2-0.5*(-106.71+120.26)*0.625*0.625*0.67 = 15.28 kN-m/m

SF_{3x} = (133.09-33)*(0.625-0.342)+0.5*(139.23-133.09)*(0.625-0.342) = 29.19 kN/m

$S.F_{(punching)}$ = (2.25-0.45)*(122.97-33) = 161.94 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	28.01	0.24	0.32	0.001	261.40	615.60	16ϕT	326.61	300	200	1005.31
At Top	0.00	0.00	0.00	0.000	0.00	615.60	16ϕT	326.61	300	200	1005.31

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 42.24 kN/m

Shear stress , τ_v= (42.24*1000)/(1000*342)

= 0.12 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 183.49 kN

Strength of Concrete in shear Vc1 = 4194.43 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 2065.11 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 1823.67 kN **Table 22.6.5.2_ (c)**

Design Vc= 1823.67 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (183.49*1000)/(2704*342)

= 0.20 N/mm^2 **No shear reinforcement reqd.**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions						
24 1.4DL	125.47	120.47	129.09	13.28	11.52	19.34
25 1.4D+1.4T+0.5LR	128.94	125.69	131.29	13.57	12.42	19.70
26 1.4D+1.4T+0.5LR	128.94	125.69	131.29	13.57	12.42	19.70
27 1.2D+0.8WX1+1.6LR	135.36	121.73	145.24	15.58	10.76	22.94
28 1.2D+1.6WZ1+0.5LR	139.75	75.25	186.48	21.66	-1.13	33.38
29 1.2D+1.6WX1+0.5LR	131.44	126.39	135.10	14.11	12.33	20.54
30 1.2D+1.6WZ1+0.5LR	139.75	75.25	186.48	21.66	-1.13	33.38
31 1.2D+EX	117.96	112.43	121.97	12.31	10.35	17.95
32 1.2D+EZ	110.84	97.84	120.25	12.13	7.54	17.94
33 0.9D+1.6WX1	110.48	102.29	116.42	11.56	8.67	16.96
34 0.9D+1.6WZ1	123.50	66.50	164.79	18.61	-1.52	28.73
35 0.9D+EZ	91.71	78.56	101.24	9.51	4.86	14.16
36 0.9D+EX	98.23	93.76	101.46	9.47	7.89	13.82
47 1.2D+0.8WX2+1.6LR	119.40	106.49	128.76	13.30	8.74	19.62
48 1.2D+1.6WZ2+0.5LR	167.46	102.42	214.59	25.54	2.56	39.00
49 1.2D+1.6WX2+0.5LR	99.13	96.34	101.15	9.41	8.43	13.68
53 0.9D+1.6WX2	75.85	74.54	76.81	6.05	5.58	8.78
54 0.9D+1.6WZ2	149.67	84.20	197.11	23.13	0.00	35.54

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Sample calculation of BM & SF

Load case : 24 1.4DL

p_{1z} = 131.3-(131.3-114.64)*0.525/1.5= 125.47 kN/m^2

p_{2z} = 114.64+(131.3-114.64)*0.525/1.5= 120.47 kN/m^2

p_{3z} = 131.3-(131.3-114.64)*(0.525-0.326)/1.5= 129.09 kN/m^2

Overhang = 0.525 m

Effective depth = 0.326 m

BM_{1z} = (125.47-33)*0.525*0.525/2+0.5*(131.3-125.47)*0.525*0.525*0.67 = 13.28 kN-m/m

BM_{2z} = (120.47-33)*0.525*0.525/2-0.5*(-114.64+120.47)*0.525*0.525*0.67 = 11.52 kN-m/m

SF_{3z} = (129.09-33)*(0.525-0.326)+0.5*(131.3-129.09)*(0.525-0.326) = 19.34 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	25.54	0.240	0.320	0.0008	262.31	586.80	16ϕT	342.64	300	200	1005.3
Top	-1.52	-0.014	-0.019	0.0000	-0.15	586.80	16ϕT	342.64	300	200	1005.3

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 39.00 kN/m

Shear stress , τ_v= (39*1000)/(1000*326)

= 0.12 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **16ϕT** **200** c/c

At Top **16ϕT** **200** c/c

Along Z Direction :

At Bottom **16ϕT** **200** c/c

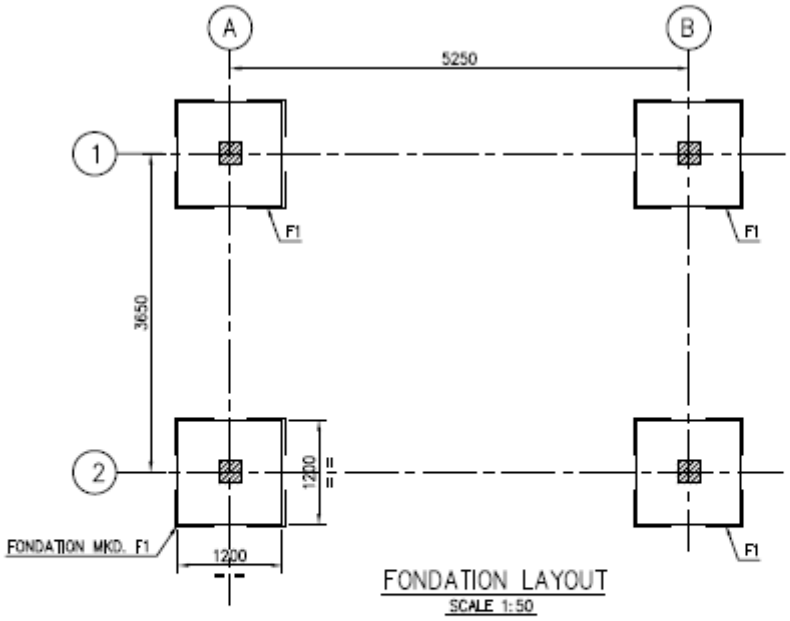
At Top **16ϕT** **200** c/c

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13.2. Foundation Design Calculation_ Compressor Room



FOOTING MARK	LEAN CONCRETE			REINFORCED CONCRETE			BOTTOM REINFORCEMENT		TOP REINFORCEMENT		Ha	REMARKS
	L	W	T	L	W	T	SHORT BARS	LONG BARS	SHORT BARS	LONG BARS		
F1	1400	1400	50	1200	1200	400	ø16@200	ø16@200	–	–	1200	

Foundation Reinforcement Summary

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13.2.1 DESIGN CALCULATION FOR COMPRESSOR ROOM FOOTING TYPE F1

	Along-X	Along-Z
Size of column :	0.25m	0.25m
Size of fdn. raft :	1.00m	1.00m
Thickness of raft :	0.40	m
Original ground level:	0.00	m
Finished floor level :	0.50	m
U/S level of foundation :	-1.20	m
U/S level Base Plate :	0.50	m
Unit weight of conc :	24	kN/m3
Unit weight of soil :	18	kN/m3

SYMBOLS AND NOTATIONS

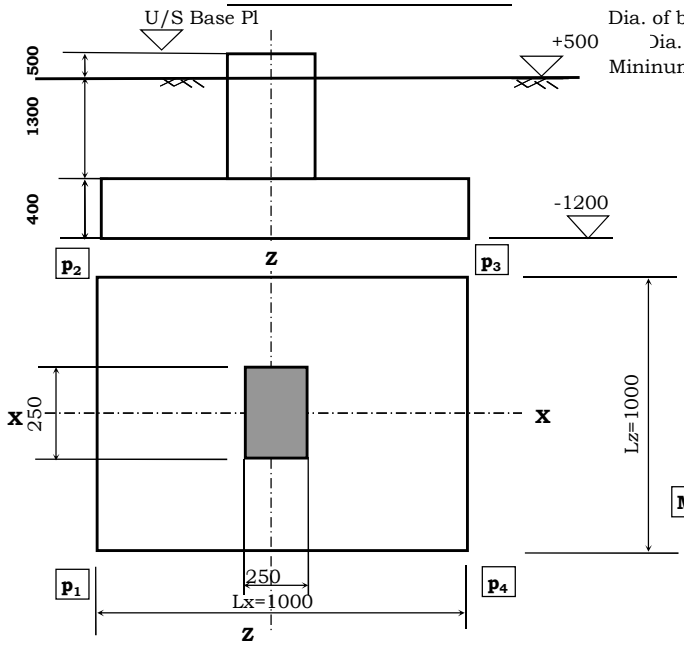
Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

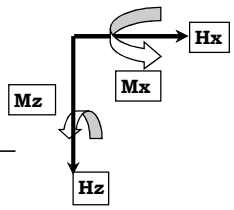
Calculation of Weight of Foundation :

ght of raft & Col.= (1*1*0.4*24)+(0.25*0.25*1.3*24) = 11.55KN
Weight of soil = ((1*1)-(0.25*0.25))*1.3*18 = 21.94KN
Total = 33.49KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%





Effective depth of slab along X direction = 342mm = 0.342m
Effective depth of slab along Z direction = 326mm = 0.326m
Clear projection along Z direction = 0.375m
Clear projection along X direction = 0.375m

H_x=Force along X-X direction
Hz=Force along Z-Z direction
M_x=Moment about X-X direction
Mz=Moment about Z-Z direction

PROPRIETARY & CONFIDENTIAL

contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part all of the contents in any form without the prior written permission of Outotec is prohibited.

document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either full or in part by such person or body without our prior permission in writing.

13.2.1 DESIGN CALCULATION FOR COMPRESSOR ROOM FOOTING TYPE F1

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design										
10 DL	148.9	-2.2	1.2	5.9	3.3	182.43	-2.23	1.22	5.91	3.26
11 DL+T	148.9	-2.2	1.2	5.9	3.3	182.43	-2.23	1.22	5.91	3.26
12 DL+LR	151.4	-2.6	1.5	5.6	3.0	184.93	-2.60	1.54	5.56	2.95
13 D+0.75LR	150.8	-2.5	1.5	5.6	3.0	184.31	-2.50	1.46	5.65	3.03
14 D+0.7EZ	146.7	-2.2	1.2	4.5	2.1	180.23	-2.23	1.22	4.54	2.07
15 D+0.7EX	151.3	-3.3	2.2	5.8	3.2	184.78	-3.33	2.19	5.83	3.19
16 D+0.7WX1+0.75LR	164.0	-11.4	9.2	5.1	2.6	197.48	-11.39	9.25	5.12	2.56
17 D+0.75WZ1+0.75LR	135.4	-2.4	1.4	-7.2	-8.2	168.91	-2.42	1.38	-7.23	-8.16
18 0.6D+WX1	106.9	-13.2	11.1	2.8	1.3	140.42	-13.18	11.12	2.84	1.33
19 0.6D+WZ1	68.8	-1.2	0.6	-13.6	-13.0	102.32	-1.23	0.63	-13.62	-12.96
20 0.6D+0.7EX	91.7	-2.4	1.7	3.5	1.9	125.20	-2.44	1.71	3.47	1.89
21 0.6D+0.7EZ	87.2	-1.3	0.7	2.2	0.8	120.65	-1.34	0.73	2.18	0.77
22 D+WX1	166.5	-14.1	11.6	5.2	2.6	200.00	-14.07	11.60	5.21	2.63
23 D+WZ1	128.4	-2.1	1.1	-11.3	-11.7	161.89	-2.12	1.12	-11.26	-11.65
39 D+0.75WX2+0.75LR	137.6	6.4	-6.3	6.2	3.5	171.11	6.39	-6.34	6.22	3.54
40 D+0.75WZ2+0.75LR	166.5	-2.6	1.6	18.6	14.3	199.98	-2.64	1.58	18.61	14.29
41 0.6D+WX2	71.8	10.5	-9.7	4.3	2.6	105.25	10.53	-9.67	4.31	2.64
42 0.6D+WZ2	110.3	-1.5	0.9	20.8	17.0	143.75	-1.51	0.90	20.83	16.98
45 D+WX2	131.3	9.6	-9.2	6.7	3.9	164.83	9.63	-9.18	6.68	3.94
46 D+WZ2	169.8	-2.4	1.4	23.2	18.3	203.33	-2.40	1.38	23.19	18.28

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Strength Design										
24 1.4DL	208.5	-3.1	1.7	8.3	4.6	242.00	-3.12	1.71	8.27	4.57
25 1.4D+1.4T+0.5LR	209.7	-3.2	1.8	7.9	4.3	243.17	-3.22	1.82	7.89	4.29
26 1.4D+1.4T+0.5LR	209.7	-3.2	1.8	7.9	4.3	243.17	-3.22	1.82	7.89	4.29
27 1.2D+0.8WX1+1.6LR	196.8	-12.7	10.3	6.0	2.9	230.28	-12.74	10.28	5.97	2.91
28 1.2D+1.6WZ1+0.5LR	147.1	-2.7	1.5	-20.6	-20.1	180.61	-2.69	1.46	-20.55	-20.11
29 1.2D+1.6WX1+0.5LR	208.1	-21.8	18.2	5.8	2.8	241.58	-21.81	18.24	5.79	2.75
30 1.2D+1.6WZ1+0.5LR	147.1	-2.7	1.5	-20.6	-20.1	180.61	-2.69	1.46	-20.55	-20.11
31 1.2D+EX	182.1	-4.2	2.9	7.0	3.8	215.57	-4.24	2.85	6.98	3.81
32 1.2D+EZ	175.6	-2.7	1.5	5.1	2.2	209.07	-2.68	1.46	5.14	2.21
33 0.9D+1.6WX1	162.2	-21.0	17.7	4.2	1.9	195.65	-20.95	17.71	4.20	1.93
34 0.9D+1.6WZ1	101.2	-1.8	0.9	-22.1	-20.9	134.68	-1.84	0.94	-22.15	-20.93
35 0.9D+EZ	130.9	-2.0	1.1	3.4	1.2	164.39	-2.01	1.10	3.37	1.24
36 0.9D+EX	137.4	-3.6	2.5	5.2	2.8	170.89	-3.57	2.49	5.21	2.84
47 1.2D+0.8WX2+1.6LR	168.7	6.2	-6.3	7.1	4.0	202.14	6.23	-6.34	7.14	3.96
48 1.2D+1.6WZ2+0.5LR	213.4	-3.1	1.9	34.6	27.8	246.91	-3.14	1.89	34.57	27.79
49 1.2D+1.6WX2+0.5LR	151.8	16.1	-15.0	8.1	4.9	185.31	16.12	-15.02	8.14	4.85
53 0.9D+1.6WX2	105.9	17.0	-15.5	6.5	4.0	139.38	16.97	-15.54	6.54	4.03
54 0.9D+1.6WZ2	167.5	-2.3	1.4	33.0	27.0	200.97	-2.29	1.36	32.97	26.96

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Check for overturning and sliding:

						Overturning		Sliding	
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
10 DL	182.4	-2.2	1.2	5.9	3.3	74.8	28.0	32.8	12.3
11 DL+T	182.4	-2.2	1.2	5.9	3.3	74.8	28.0	32.8	12.3
12 DL+LR	184.9	-2.6	1.5	5.6	3.0	60.0	31.3	28.5	13.3
13 D+0.75LR	184.3	-2.5	1.5	5.6	3.0	63.1	30.4	29.4	13.1
14 D+0.7EZ	180.2	-2.2	1.2	4.5	2.1	73.9	43.5	32.3	15.9
15 D+0.7EX	184.8	-3.3	2.2	5.8	3.2	42.1	28.9	22.2	12.7
16 D+0.7WX1+0.75LR	197.5	-11.4	9.2	5.1	2.6	10.7	38.6	6.9	15.4
17 D+0.75WZ1+0.75LR	168.9	-2.4	1.4	-7.2	-8.2	61.0	10.4	27.9	9.3
18 0.6D+WX1	140.4	-13.2	11.1	2.8	1.3	6.3	52.8	4.3	19.7
19 0.6D+WZ1	102.3	-1.2	0.6	-13.6	-13.0	81.2	3.9	33.3	3.0
20 0.6D+0.7EX	125.2	-2.4	1.7	3.5	1.9	36.7	33.2	20.6	14.4
21 0.6D+0.7EZ	120.7	-1.3	0.7	2.2	0.8	82.4	78.8	36.1	22.1
22 D+WX1	200.0	-14.1	11.6	5.2	2.6	8.6	38.0	5.7	15.4
23 D+WZ1	161.9	-2.1	1.1	-11.3	-11.7	72.4	6.9	30.5	5.8
39 D+0.75WX2+0.75LR	171.1	6.4	-6.3	6.2	3.5	13.5	24.2	10.7	11.0
40 D+0.75WZ2+0.75LR	200.0	-2.6	1.6	18.6	14.3	63.2	7.0	30.3	4.3
41 0.6D+WX2	105.3	10.5	-9.7	4.3	2.6	5.4	19.9	4.0	9.8
42 0.6D+WZ2	143.8	-1.5	0.9	20.8	17.0	80.3	4.2	38.0	2.8
45 D+WX2	164.8	9.6	-9.2	6.7	3.9	9.0	20.9	6.8	9.9
46 D+WZ2	203.3	-2.4	1.4	23.2	18.3	73.5	5.6	33.8	3.5
					Min.=	5.4	3.9	4.0	2.8
						OK	OK	OK	OK

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13.2.1 DESIGN CALCULATION FOR COMPRESSOR ROOM FOOTING TYPE F1

Load Combination & Calculation of Base pressure

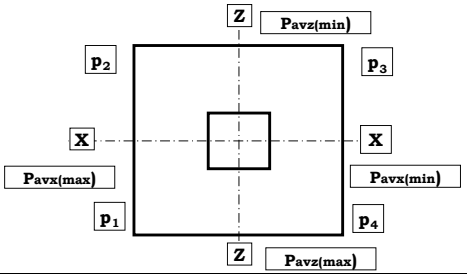
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 350 kN/m^2 For WL/SL case = 437.5 kN/m^2

Gross SBC of soil for DL, LL= 371.6 kN/m^2 For WL/SL case = 459.1 kN/m^2



Allowable Stress Design																	
Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
	V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
10 DL	182.43	1.22	3.26	182.43	7.32	19.57	209.32	170.18	194.68	155.54	187.72	189.75	175.11	202.00	162.86	O.K
11 DL+T	182.43	1.22	3.26	182.43	7.32	19.57	209.32	170.18	194.68	155.54	187.72	189.75	175.11	202.00	162.86	O.K
12 DL+LR	184.93	1.54	2.95	184.93	9.25	17.71	211.88	176.47	193.39	157.98	190.28	194.18	175.69	202.64	167.23	O.K
13 D+0.75LR	184.31	1.46	3.03	184.31	8.77	18.17	211.25	174.90	193.71	157.37	189.65	193.08	175.54	202.48	166.14	O.K
14 D+0.7EZ	180.23	1.22	2.07	180.23	7.32	12.43	199.97	175.12	185.33	160.48	178.37	187.55	172.91	192.65	167.80	O.K
15 D+0.7EX	184.78	2.19	3.19	184.78	13.16	19.15	217.09	178.78	190.77	152.47	195.49	197.94	171.62	203.93	165.63	O.K
16 D+0.7WX1+0.75LR	197.48	9.25	2.56	197.48	55.48	15.35	268.31	237.62	157.35	126.65	246.71	252.97	142.00	212.83	182.14	O.K
17 D+0.75WZ1+0.75LR	168.91	1.38	-8.16	168.91	8.30	48.95	226.16	128.26	209.55	111.65	204.56	177.21	160.60	217.86	119.96	O.K
18 0.6D+WX1	140.42	11.12	1.33	140.42	66.69	7.97	215.09	199.14	81.71	65.76	193.49	207.12	73.74	148.40	132.45	O.K
19 0.6D+WZ1	102.32	0.63	-12.96	102.32	3.78	77.75	183.85	28.34	176.29	20.78	162.25	106.10	98.54	180.07	24.56	O.K
20 0.6D+0.7EX	125.20	1.71	1.89	125.20	10.23	11.32	146.75	124.11	126.29	103.65	125.15	135.43	114.97	136.52	113.88	O.K
21 0.6D+0.7EZ	120.65	0.73	0.77	120.65	4.39	4.60	129.64	120.45	120.86	111.66	108.04	125.05	116.26	125.25	116.06	O.K
22 D+WX1	200.00	11.60	2.63	200.00	69.62	15.80	285.42	253.81	146.18	114.58	263.82	269.62	130.38	215.80	184.20	O.K
23 D+WZ1	161.89	1.12	-11.65	161.89	6.71	69.92	238.52	98.68	225.11	85.26	216.92	168.60	155.19	231.82	91.97	O.K
39 D+0.75WX2+0.75LR	171.11	-6.34	3.54	171.11	38.04	21.24	230.39	187.91	154.31	111.83	208.79	209.15	133.07	192.35	149.87	O.K
40 D+0.75WZ2+0.75LR	199.98	1.58	14.29	199.98	9.50	85.75	295.23	123.73	276.23	104.73	273.63	209.48	190.48	285.73	114.23	O.K
41 0.6D+WX2	105.25	-9.67	2.64	105.25	58.01	15.83	179.10	147.43	63.08	31.41	157.50	163.27	47.25	121.09	89.42	O.K
42 0.6D+WZ2	143.75	0.90	16.98	143.75	5.37	101.85	250.97	47.27	240.23	36.53	229.37	149.12	138.38	245.60	41.90	O.K
45 D+WX2	164.83	-9.18	3.94	164.83	55.08	23.66	243.57	196.24	133.41	86.08	221.97	219.91	109.75	188.49	141.16	O.K
46 D+WZ2	203.33											211.62	195.03	313.01	93.65	O.K

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Strength Design

24 1.4DL	242.00	1.71	4.57	242.00	10.25	27.40	279.65	224.85	259.16	204.35	258.05	252.25	231.76	269.41	214.60	
25 1.4D+1.4T+0.5LR	243.17	1.82	4.29	243.17	10.91	25.76	279.84	228.32	258.02	206.51	258.24	254.08	232.27	268.93	217.42	
26 1.4D+1.4T+0.5LR	243.17	1.82	4.29	243.17	10.91	25.76	279.84	228.32	258.02	206.51	258.24	254.08	232.27	268.93	217.42	
27 1.2D+0.8WX1+1.6LR	230.28	10.28	2.91	230.28	61.70	17.48	309.46	274.49	186.07	151.10	287.86	291.98	168.59	247.77	212.80	
28 1.2D+1.6WZ1+0.5LR	180.61	1.46	-20.11	180.61	8.77	120.64	310.01	68.74	292.48	51.21	288.41	189.38	171.85	301.25	59.98	
29 1.2D+1.6WX1+0.5LR	241.58	18.24	2.75	241.58	109.42	16.52	367.52	334.47	148.69	115.64	345.92	351.00	132.17	258.11	225.06	
30 1.2D+1.6WZ1+0.5LR	180.61	1.46	-20.11	180.61	8.77	120.64	310.01	68.74	292.48	51.21	288.41	189.38	171.85	301.25	59.98	
31 1.2D+EX	215.57	2.85	3.81	215.57	17.12	22.88	255.58	209.81	221.33	175.56	233.98	232.70	198.45	238.46	192.69	
32 1.2D+EZ	209.07	1.46	2.21	209.07	8.78	13.28	231.13	204.58	213.57	187.01	209.53	217.86	200.29	222.35	195.80	
33 0.9D+1.6WX1	195.65	17.71	1.93	195.65	106.26	11.58	313.49	290.33	100.97	77.81	291.89	301.91	89.39	207.23	184.07	
34 0.9D+1.6WZ1	134.68	0.94	-20.93	134.68	5.61	125.58	265.87	14.71	254.65	3.49	244.27	140.29	129.07	260.26	9.10	
35 0.9D+EZ	164.39	1.10	1.24	164.39	6.59	7.41	178.39	163.57	165.21	150.39	156.79	170.98	157.80	171.80	156.98	
36 0.9D+EX	170.89	2.49	2.84	170.89	14.93	17.02	202.83	168.80	172.98	138.94	181.23	185.82	155.96	187.91	153.87	
47 1.2D+0.8WX2+1.6LR	202.14	-6.34	3.96	202.14	38.06	23.78	263.98	216.42	187.86	140.31	242.38	240.20	164.09	225.92	178.37	
48 1.2D+1.6WZ2+0.5LR	246.91	1.89	27.79	246.91	11.31	166.72	424.94	91.49	402.32	68.87	403.34	258.22	235.60	413.63	80.18	
49 1.2D+1.6WX2+0.5LR	185.31	-15.02	4.85	185.31	90.10	29.10	304.51	246.31	124.31	66.11	282.91	275.41	95.21	214.41	156.21	
53 0.9D+1.6WX2	139.38	-15.54	4.03	139.38	93.25	24.16	256.79	208.47	70.29	21.96	235.19	232.63	46.13	163.54	115.22	
54 0.9D+1.6WZ2	200.97	1.36	26.96	200.97	8.15	161.78	370.91	47.34	354.60	31.03	349.31	209.13	192.82	362.76	39.19	
Max :	247	18	28													
Min :	102	-16	-21													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 190.28 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 200.28 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 299.70 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 309.70 kN/m²

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Sample calculation of values of base pressure:

Load case : 11 DL+T

Maximum pressure, p_1 (pmax) = 182.43+7.32+19.57= 209.32 kN/m^2

p_2= 182.43+7.32-19.57= 170.18 kN/m^2

p_3= 182.43-(7.32+19.57)= 155.54 kN/m^2

p_4= 182.43-7.32+19.57= 194.68 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 209.32-1.2*18= 187.72 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (209.32+170.18)/2= 189.75 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (155.54+194.68)/2= 175.11 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (209.32+194.68)/2= 202.00 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (155.54+170.18)/2= 162.86 kN/m^2

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13.2.1 DESIGN CALCULATION FOR COMPRESSOR ROOM FOOTING TYPE F1

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.25+2*0.5*0.326)+(0.25+2*0.5*0.326))*2

= 2.304 m

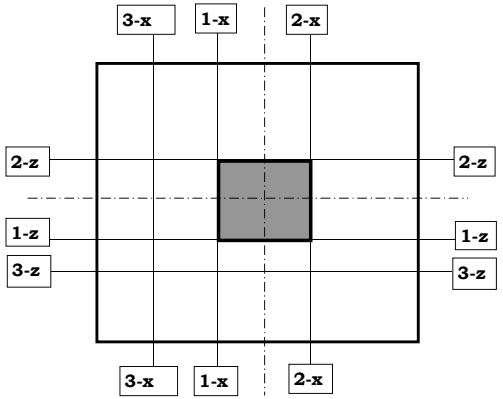
Punching Area= ((0.25+2*0.5*0.326)*(0.25+2*0.5*0.326))

= 0.33 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.3*18)

= 33.00 kN/m^2

Area of footing, A_f= 1.00 m^2



Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions							
24 1.4DL	244.57	239.44	251.57	15.24	14.16	7.22	140.03
25 1.4D+1.4T+0.5LR	245.90	240.45	253.36	15.35	14.20	7.28	140.82
26 1.4D+1.4T+0.5LR	245.90	240.45	253.36	15.35	14.20	7.28	140.82
27 1.2D+0.8WX1+1.6LR	245.71	214.86	287.91	17.13	10.62	8.48	132.18
28 1.2D+1.6WZ1+0.5LR	182.81	178.42	188.80	10.84	9.92	5.15	98.90
29 1.2D+1.6WX1+0.5LR	268.94	214.23	343.78	20.44	8.90	10.37	139.75
30 1.2D+1.6WZ1+0.5LR	182.81	178.42	188.80	10.84	9.92	5.15	98.90
31 1.2D+EX	219.86	211.29	231.57	13.74	11.93	6.57	122.32
32 1.2D+EZ	211.27	206.88	217.28	12.84	11.92	6.09	117.97
33 0.9D+1.6WX1	222.22	169.09	294.90	17.04	5.83	8.76	108.97
34 0.9D+1.6WZ1	136.08	133.28	139.92	7.45	6.85	3.53	68.12
35 0.9D+EZ	166.04	162.74	170.55	9.59	8.89	4.55	88.03
36 0.9D+EX	174.62	167.16	184.83	10.48	8.91	5.03	92.38
47 1.2D+0.8WX2+1.6LR	211.66	192.63	237.69	13.90	9.89	6.80	113.33
48 1.2D+1.6WZ2+0.5LR	249.74	244.08	257.47	15.64	14.44	7.42	143.32
49 1.2D+1.6WX2+0.5LR	207.84	162.79	269.46	15.46	5.96	7.90	102.05
53 0.9D+1.6WX2	162.69	116.07	226.48	12.40	2.56	6.49	71.27
54 0.9D+1.6WZ2	203.01	198.94	208.59	12.24	11.38	5.80	112.54

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Sample calculation of BM & SF

Load case : 24 1.4DL

p_{1x} = 252.25-(252.25-231.76)*0.375/1 = 244.57 kN/m² Overhang = 0.375 m

p_{2x} = 231.76+(252.25-231.76)*0.375/1 = 239.44 kN/m² Effective depth = 0.342 m

p_{3x} = 252.25-(252.25-231.76)*(0.375-0.342)/1 251.57 kN/m²

BM_{1x} = (244.57-33)*0.375*0.375/2+0.5*(252.25-244.57)*0.375*0.375*0.67 = 15.24 kN-m/m

BM_{2x} = (239.44-33)*0.375*0.375/2-0.5*(-231.76+239.44)*0.375*0.375*0.67 = 14.16 kN-m/m

SF_{3x} = (251.57-33)*(0.375-0.342)+0.5*(252.25-251.57)*(0.375-0.342) = 7.22 kN/m

$S.F_{(punching)}$ = (1-0.33)*(242.00-33) = 140.03 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	20.44	0.17	0.23	0.001	190.45	615.60	16ϕT	326.61	300	200	1005.31
At Top	0.00	0.00	0.00	0.000	0.00	615.60	16ϕT	326.61	300	200	1005.31

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 10.37 kN/m

Shear stress , τ_v= (10.37*1000)/(1000*342)

= 0.03 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 143.32 kN

Strength of Concrete in shear Vc1 = 2330.84 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1930.23 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 1553.89 kN **Table 22.6.5.2_ (c)**

Design Vc= 1553.89 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (143.32*1000)/(2304*342)

= 0.18 N/mm^2 **No shear reinforcement reqd.**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions						
24 1.4DL	248.86	235.15	266.72	16.15	13.25	11.52
25 1.4D+1.4T+0.5LR	249.61	236.74	266.41	16.14	13.42	11.50
26 1.4D+1.4T+0.5LR	249.61	236.74	266.41	16.14	13.42	11.50
27 1.2D+0.8WX1+1.6LR	234.66	225.91	246.06	14.80	12.95	10.48
28 1.2D+1.6WZ1+0.5LR	210.77	150.46	289.43	16.76	4.00	12.85
29 1.2D+1.6WX1+0.5LR	245.72	237.45	256.49	15.54	13.79	10.99
30 1.2D+1.6WZ1+0.5LR	210.77	150.46	289.43	16.76	4.00	12.85
31 1.2D+EX	221.30	209.85	236.22	14.05	11.63	10.01
32 1.2D+EZ	212.39	205.76	221.05	13.08	11.68	9.25
33 0.9D+1.6WX1	198.55	192.76	206.10	12.05	10.82	8.51
34 0.9D+1.6WZ1	166.08	103.29	247.95	13.79	0.50	10.83
35 0.9D+EZ	166.24	162.54	171.07	9.63	8.85	6.78
36 0.9D+EX	175.15	166.64	186.24	10.60	8.79	7.55
47 1.2D+0.8WX2+1.6LR	208.09	196.20	223.59	13.15	10.64	9.40
48 1.2D+1.6WZ2+0.5LR	288.59	205.22	397.29	23.86	6.22	18.25
49 1.2D+1.6WX2+0.5LR	192.59	178.04	211.56	12.25	9.17	8.82
53 0.9D+1.6WX2	145.42	133.34	161.17	8.76	6.20	6.34
54 0.9D+1.6WZ2	241.42	160.53	346.91	20.37	3.25	15.77

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Sample calculation of BM & SF

Load case : 24 1.4DL

p_{1z} = 269.41-(269.41-214.6)*0.375/1= 248.86 kN/m^2

p_{2z} = 214.6+(269.41-214.6)*0.375/1= 235.15 kN/m^2

p_{3z} = 269.41-(269.41-214.6)*(0.375-0.326)/1= 266.72 kN/m^2

Overhang = 0.375 m

Effective depth = 0.326 m

BM_{1z} = (248.86-33)*0.375*0.375/2+0.5*(269.41-248.86)*0.375*0.375*0.67 = 16.15 kN-m/m

BM_{2z} = (235.15-33)*0.375*0.375/2-0.5*(-214.6+235.15)*0.375*0.375*0.67 = 13.25 kN-m/m

SF_{3z} = (266.72-33)*(0.375-0.326)+0.5*(269.41-266.72)*(0.375-0.326) = 11.52 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	23.86	0.225	0.299	0.0007	245.01	586.80	16ϕT	342.64	300	200	1005.3
Top	0.00	0.000	0.000	0.0000	0.00	586.80	16ϕT	342.64	300	200	1005.3

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 18.25 kN/m

Shear stress , τ_v= (18.25*1000)/(1000*326)

= 0.06 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **16ϕT** **200** c/c

At Top **16ϕT** **200** c/c

Along Z Direction :

At Bottom **16ϕT** **200** c/c

At Top **16ϕT** **200** c/c

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

REFERENCES

1.) OVERALL LAYOUT PLAN — MD-216-49FE-ME-LAD-1007

2.) LAYOUT & DETAILS OF FDN, COL &GRADE BEAM – COMPRESSOR ROOM — MD-216-49FE-EG-CV-GAD-1626

3.) LAYOUT & DETAILS OF FDN, COL &GRADE BEAM – VAPORIZER ROOM — MD-216-49FE-EG-CV-GAD-1626

4.) GEOTECHNICAL REPORT — MD-216-3000-EG-CV-RPT-1008

5.) STRUCTURAL DESIGN CRITERIA — MD-216-0000-EG-ST-CRT-0001

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ATTACHMENT 1 STAAD INPUT DATA_ VAPORIZER ROOM

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

STAAD INPUT:

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 12-Oct-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -1.7 0; 2 5.25 -1.7 0; 3 0 -1.7 8.25; 4 5.25 -1.7 8.25; 5 0 5.035 0;
6 5.25 5.035 0; 7 0 5.035 8.25; 8 5.25 5.035 8.25; 9 5.25 0 0; 10 0 0 8.25;
11 5.25 0 8.25; 12 0 0 0; 13 0 -1.7 4.3; 14 0 0 4.3; 15 5.25 -1.7 4.3;
16 5.25 0 4.3; 17 5.25 5.035 4.3; 18 0 5.035 4.3; 19 5.25 0 6.05;
20 5.25 5.035 6.05; 21 5.25 0 1.95; 22 5.25 5.035 1.95; 23 0 5.035 1.95;
24 0 5.035 6.05; 25 1.85 5.035 4.3; 26 1.85 5.035 1.95; 27 1.85 5.035 6.05;
28 1.85 5.035 0; 29 3.55 5.035 1.95; 30 3.55 5.035 4.3; 31 3.55 5.035 6.05;
32 3.55 5.035 0;
MEMBER INCIDENCES
1 1 12; 2 2 9; 3 3 10; 4 4 11; 5 9 6; 6 10 7; 7 11 8; 8 12 5; 9 12 9; 10 12 14;
11 9 21; 12 10 11; 13 5 28; 14 5 23; 15 6 22; 16 7 8; 17 14 10; 18 13 14;
19 16 19; 20 15 16; 21 17 20; 22 16 17; 23 18 24; 24 14 18; 25 18 25; 26 19 11;
27 20 8; 28 19 20; 29 21 16; 30 22 17; 31 21 22; 32 23 18; 33 24 7; 34 22 29;
35 20 31; 36 25 30; 37 26 23; 38 27 24; 39 26 25; 40 25 27; 41 28 32; 42 26 28;
43 29 26; 44 30 17; 45 29 30; 46 31 27; 47 30 31; 48 32 6; 49 29 32;
DEFINE MATERIAL START
ISOTROPIC CONCRETE_M35
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
G 9.28139e+006
TYPE CONCRETE
STRENGTH FCU 28000
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 3 4 6 TO 8 18 20 22 24 28 31 PRIS YD 0.25 ZD 0.25
2 5 PRIS YD 0.25 ZD 0.45
13 TO 16 21 23 27 30 32 33 41 48 PRIS YD 0.5 ZD 0.25
9 TO 12 17 19 26 29 PRIS YD 0.7 ZD 0.25
MEMBER PROPERTY AMERICAN
25 34 TO 40 42 TO 47 49 PRIS YD 0.4 ZD 0.25
CONSTANTS
MATERIAL CONCRETE_M35 ALL
SUPPORTS
1 TO 4 13 15 FIXED
MEMBER CRACKED
1 TO 8 18 20 22 24 28 31 REDUCTION RIX 0.700000 RIY 0.700000 RIZ 0.700000
9 TO 17 19 21 23 25 TO 27 29 30 32 TO 48 -
49 REDUCTION RIX 0.350000 RIY 0.350000 RIZ 0.350000
DEFINE IBC 2003
SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 4
SELFWEIGHT 1
MEMBER WEIGHT
14 15 21 23 27 30 32 33 UNI 15.3
```

9 10 12 17 26 29 UNI 17.2
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.300 M 5.250 M 8.250 M 2.000 0.010 0 -
0 0 0 0 1.057 1.000 1.000 0.850 0 -
0 0 0 0.893 0.800 -0.180
!> END GENERATED DATA BLOCK
INT 1.16718 1.16718 1.16915 1.17109 1.17302 1.17493 1.17682 1.17869 1.18054 -
 1.18238 1.1842 1.186 1.18779 1.18956 1.19132 HEIG 0 4.572 4.628 4.684 -
 4.74 4.796 4.852 4.908 4.964 5.02 5.076 5.132 5.188 5.244 5.3
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.300 M 5.250 M 8.250 M 2.000 0.010 1 -
0 0 0 0 1.057 1.000 1.000 0.850 0 -
0 0 0 0.893 -0.500 0.180
!> END GENERATED DATA BLOCK
INT -0.834496 -0.834496 HEIG 0 5.3
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.300 M 5.250 M 8.250 M 2.000 0.010 2 -
0 0 0 0 1.057 1.000 1.000 0.850 0 -
0 0 0 0.893 -0.700 0.180
!> END GENERATED DATA BLOCK
INT -1.07238 -1.07238 HEIG 0 5.3
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 MPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.300 M 8.250 M 5.250 M 2.000 0.010 0 -
0 0 0 0 1.057 1.000 1.000 0.850 0 -
0 0 0 0.897 0.800 -0.180
!> END GENERATED DATA BLOCK
INT 3.03473 3.03473 3.03984 3.04491 3.04992 3.05489 3.05981 3.06468 3.06951 -
 3.07429 3.07902 3.08372 3.08837 3.09298 3.09755 HEIG 0 4.572 4.628 4.684 -
 4.74 4.796 4.852 4.908 4.964 5.02 5.076 5.132 5.188 5.244 5.3
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.300 M 8.250 M 5.250 M 2.000 0.010 1 -
0 0 0 0 1.057 1.000 1.000 0.850 0 -
0 0 0 0.897 -0.386 0.180
!> END GENERATED DATA BLOCK
INT -0.700806 -0.700806 HEIG 0 5.3
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.300 M 8.250 M 5.250 M 2.000 0.010 2 -
0 0 0 0 1.057 1.000 1.000 0.850 0 -
0 0 0 0.897 -0.700 0.180
!> END GENERATED DATA BLOCK
INT -1.07645 -1.07645 HEIG 0 5.3

```
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
14 15 21 23 27 30 32 33 UNI GY -15.3
9 10 12 17 26 29 UNI GY -17.2
LOAD 4 LOADTYPE Live REDUCIBLE TITLE LL
MEMBER LOAD
14 15 21 23 27 30 32 33 UNI GY -1.3125
15 CON GY -25 0.975
15 CON GZ 5 0.975
15 CON GX 2.5 0.975
21 CON GY -25 0.875
21 CON GZ 5 0.875
21 CON GZ 2.5 0.875
LOAD 5 LOADTYPE Wind  TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 6
WIND LOAD -X 1 TYPE 2 YR 0 6
WIND LOAD -Z 1 TYPE 3 YR 0 6
WIND LOAD -Z -1 TYPE 3 YR 0 6
LOAD 6 LOADTYPE Wind  TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 6
WIND LOAD -Z 1 TYPE 5 YR 0 6
WIND LOAD -X 1 TYPE 6 YR 0 6
WIND LOAD -X -1 TYPE 6 YR 0 6
LOAD 7 LOADTYPE Wind  TITLE WL-X
WIND LOAD X -1 TYPE 1 YR 0 6
WIND LOAD -X -1 TYPE 2 YR 0 6
WIND LOAD -Z 1 TYPE 3 YR 0 6
WIND LOAD -Z -1 TYPE 3 YR 0 6
LOAD 8 LOADTYPE Wind  TITLE WL-Z
WIND LOAD Z -1 TYPE 4 YR 0 6
WIND LOAD -Z -1 TYPE 5 YR 0 6
WIND LOAD -X 1 TYPE 6 YR 0 6
WIND LOAD -X -1 TYPE 6 YR 0 6
LOAD 9 LOADTYPE Temperature  TITLE TEMP
TEMPERATURE LOAD
5 TO 8 13 TO 16 21 TO 25 27 28 30 TO 49 TEMP 25
LOAD COMB 10 DL
3 1.0
LOAD COMB 11 DL+T
3 1.0
LOAD COMB 12 DL+LR
3 1.0 4 1.0
LOAD COMB 13 D+0.75LR
3 1.0 4 0.75
LOAD COMB 14 D+0.7EZ
```

3 1.0 1 0.7
LOAD COMB 15 D+0.7EX
3 1.0 2 0.7
LOAD COMB 16 D+0.7WX1+0.75LR
3 1.0 5 0.75 4 0.75
LOAD COMB 17 D+0.75WZ1+0.75LR
3 1.0 6 0.75 4 0.75
LOAD COMB 18 0.6D+WX1
3 0.6 5 1.0
LOAD COMB 19 0.6D+WZ1
3 0.6 6 1.0
LOAD COMB 20 0.6D+0.7EX
3 0.6 2 0.7
LOAD COMB 21 0.6D+0.7EZ
3 0.6 1 0.7
LOAD COMB 22 D+WX1
3 1.0 5 1.0
LOAD COMB 23 D+WZ1
3 1.0 6 1.0
LOAD COMB 24 1.4DL
3 1.4
LOAD COMB 25 1.4D+1.4T+0.5LR
3 1.4 9 1.4 4 0.5
LOAD COMB 26 1.4D+1.4T+0.5LR
3 1.4 9 1.4 4 0.5
LOAD COMB 27 1.2D+0.8WX1+1.6LR
3 1.2 5 0.8 4 1.6
LOAD COMB 28 1.2D+1.6WZ1+0.5LR
3 1.2 4 0.5 6 1.6
LOAD COMB 29 1.2D+1.6WX1+0.5LR
3 1.2 4 0.5 5 1.6
LOAD COMB 30 1.2D+1.6WZ1+0.5LR
3 1.2 4 0.5 6 1.6
LOAD COMB 31 1.2D+EX
3 1.2 2 1.0
LOAD COMB 32 1.2D+EZ
3 1.2 1 1.0
LOAD COMB 33 0.9D+1.6WX1
3 0.9 5 1.6
LOAD COMB 34 0.9D+1.6WZ1
3 0.9 6 1.6
LOAD COMB 35 0.9D+EZ
3 0.9 1 1.0
LOAD COMB 36 0.9D+EX
3 0.9 2 1.0
LOAD COMB 39 D+0.75WX2+0.75LR
3 1.0 7 0.75 4 0.75
LOAD COMB 40 D+0.75WZ2+0.75LR
3 1.0 8 0.75 4 0.75
LOAD COMB 41 0.6D+WX2
3 0.6 7 1.0
LOAD COMB 42 0.6D+WZ2
3 0.6 8 1.0
LOAD COMB 45 D+WX2

3 1.0 7 1.0
LOAD COMB 46 D+WZ2
3 1.0 8 1.0
LOAD COMB 47 1.2D+0.8WX2+1.6LR
3 1.2 7 0.8 4 1.6
LOAD COMB 48 1.2D+1.6WZ2+0.5LR
3 1.2 8 1.6 4 0.5
LOAD COMB 49 1.2D+1.6WX2+0.5LR
3 1.2 7 1.6 4 0.5
LOAD COMB 53 0.9D+1.6WX2
3 0.9 7 1.6
LOAD COMB 54 0.9D+1.6WZ2
3 0.9 8 1.6
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
CLB 0.04 MEMB 1 TO 35
CLS 0.04 MEMB 1 TO 38 41 43 44 46 48
CLT 0.04 MEMB 1 TO 35
FC 28000 MEMB 1 TO 35
FYMAIN 414000 MEMB 1 TO 35
MINMAIN 16 MEMB 9 TO 17 19 21 23 25 TO 27 29 30 32 33 36 41 44 48
MINMAIN 25 MEMB 1 TO 8 18 20 22 24
MINSEC 10 MEMB 1 TO 38 41 43 44 46 48
DESIGN BEAM 9 TO 17 19 21 23 25 TO 27 29 30 32 TO 49
DESIGN COLUMN 1 TO 8 18 20 22 24 28 31
END CONCRETE DESIGN
FINISH

ATTACHMENT 2 STAAD INPUT DATA_ COMPRESSOR ROOM

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

STAAD INPUT:

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 12-Oct-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -1.7 0; 2 3.65 -1.7 0; 5 0 5.035 0; 6 3.65 5.035 0; 12 0 0 0;
13 0 -1.7 5.25; 15 3.65 -1.7 5.25; 27 3.65 0 0; 28 3.65 0 5.25;
29 3.65 5.035 5.25; 30 0 5.035 5.25; 31 0 0 5.25; 32 0 5.035 1.34;
33 3.65 5.035 1.34; 34 0 5.035 2.89; 35 3.65 5.035 2.89; 36 3 0 5.25;
37 3 5.035 5.25; 38 1.25 0 5.25; 39 1.25 5.035 5.25; 40 2.125 5.035 5.25;
41 2.125 5.035 0;
MEMBER INCIDENCES
1 1 12; 2 2 27; 5 27 6; 8 12 5; 18 13 31; 20 15 28; 22 28 29; 24 31 30;
33 12 27; 34 31 38; 35 12 31; 36 27 28; 37 5 32; 38 6 33; 39 30 39; 40 5 41;
41 32 34; 42 33 35; 44 34 30; 45 35 29; 47 36 28; 48 37 29; 49 36 37; 50 38 36;
51 39 40; 52 38 39; 53 40 37; 54 41 6; 55 40 41;
DEFINE MATERIAL START
ISOTROPIC CONCRETE_M35
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
G 9.28139e+006
TYPE CONCRETE
STRENGTH FCU 28000
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 2 5 8 18 20 22 24 49 52 PRIS YD 0.25 ZD 0.25
37 TO 42 44 45 48 51 53 54 PRIS YD 0.5 ZD 0.25
33 TO 36 47 50 PRIS YD 0.7 ZD 0.25
MEMBER PROPERTY AMERICAN
55 PRIS YD 0.4 ZD 0.25
CONSTANTS
MATERIAL CONCRETE_M35 ALL
SUPPORTS
1 2 13 15 FIXED
MEMBER CRACKED
1 2 5 8 18 20 22 24 49 52 REDUCTION RIX 0.700000 RIY 0.700000 RIZ 0.700000
33 TO 42 44 45 47 48 50 51 53 -
54 REDUCTION RIX 0.350000 RIY 0.350000 RIZ 0.350000
DEFINE IBC 2003
SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 4
SELFWEIGHT 1
MEMBER WEIGHT
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.300 M 5.250 M 8.250 M 2.000 0.010 0 -
```

```
0 0 0 0 1.057 1.000 1.000 0.850 0 -
0 0 0 0.893 0.800 -0.180
!> END GENERATED DATA BLOCK
INT 1.16718 1.16718 1.16915 1.17109 1.17302 1.17493 1.17682 1.17869 1.18054 -
 1.18238 1.1842 1.186 1.18779 1.18956 1.19132 HEIG  0  4.572  4.628  4.684 -
 4.74  4.796  4.852  4.908  4.964  5.02  5.076  5.132  5.188  5.244  5.3
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.300 M 5.250 M 8.250 M 2.000 0.010 1 -
0 0 0 0 1.057 1.000 1.000 0.850 0 -
0 0 0 0.893 -0.500 0.180
!> END GENERATED DATA BLOCK
INT -0.834496 -0.834496 HEIG  0  5.3
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.300 M 5.250 M 8.250 M 2.000 0.010 2 -
0 0 0 0 1.057 1.000 1.000 0.850 0 -
0 0 0 0.893 -0.700 0.180
!> END GENERATED DATA BLOCK
INT -1.07238 -1.07238 HEIG  0  5.3
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 MPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.300 M 8.250 M 5.250 M 2.000 0.010 0 -
0 0 0 0 1.057 1.000 1.000 0.850 0 -
0 0 0 0.897 0.800 -0.180
!> END GENERATED DATA BLOCK
INT 3.03473 3.03473 3.03984 3.04491 3.04992 3.05489 3.05981 3.06468 3.06951 -
 3.07429 3.07902 3.08372 3.08837 3.09298 3.09755 HEIG  0  4.572  4.628  4.684 -
 4.74  4.796  4.852  4.908  4.964  5.02  5.076  5.132  5.188  5.244  5.3
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.300 M 8.250 M 5.250 M 2.000 0.010 1 -
0 0 0 0 1.057 1.000 1.000 0.850 0 -
0 0 0 0.897 -0.386 0.180
!> END GENERATED DATA BLOCK
INT -0.700806 -0.700806 HEIG  0  5.3
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.300 M 8.250 M 5.250 M 2.000 0.010 2 -
0 0 0 0 1.057 1.000 1.000 0.850 0 -
0 0 0 0.897 -0.700 0.180
!> END GENERATED DATA BLOCK
INT -1.07645 -1.07645 HEIG  0  5.3
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
```

```
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
37 38 41 42 44 45 UNI GY -10.7
33 TO 36 47 UNI GY -17.2
LOAD 4 LOADTYPE Live REDUCIBLE TITLE LL
MEMBER LOAD
37 38 41 42 44 45 UNI GY -0.95
JOINT LOAD
40 FY -25
40 FX 5
40 FZ 2.5
LOAD 5 LOADTYPE Wind  TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 6
WIND LOAD -X 1 TYPE 2 YR 0 6
WIND LOAD -Z 1 TYPE 3 YR 0 6
WIND LOAD -Z -1 TYPE 3 YR 0 6
LOAD 6 LOADTYPE Wind  TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 6
WIND LOAD -Z 1 TYPE 5 YR 0 6
WIND LOAD -X 1 TYPE 6 YR 0 6
WIND LOAD -X -1 TYPE 6 YR 0 6
LOAD 7 LOADTYPE Wind  TITLE WL-X
WIND LOAD X -1 TYPE 1 YR 0 6
WIND LOAD -X -1 TYPE 2 YR 0 6
WIND LOAD -Z 1 TYPE 3 YR 0 6
WIND LOAD -Z -1 TYPE 3 YR 0 6
LOAD 8 LOADTYPE Wind  TITLE WL-Z
WIND LOAD Z -1 TYPE 4 YR 0 6
WIND LOAD -Z -1 TYPE 5 YR 0 6
WIND LOAD -X 1 TYPE 6 YR 0 6
WIND LOAD -X -1 TYPE 6 YR 0 6
LOAD 9 LOADTYPE Temperature  TITLE TEMP
TEMPERATURE LOAD
5 8 22 24 37 TO 42 44 45 48 49 51 TO 55 TEMP 25
LOAD COMB 10 DL
3 1.0
LOAD COMB 11 DL+T
3 1.0
LOAD COMB 12 DL+LR
3 1.0 4 1.0
LOAD COMB 13 D+0.75LR
3 1.0 4 0.75
LOAD COMB 14 D+0.7EZ
3 1.0 1 0.7
LOAD COMB 15 D+0.7EX
3 1.0 2 0.7
LOAD COMB 16 D+0.7WX1+0.75LR
3 1.0 5 0.75 4 0.75
LOAD COMB 17 D+0.75WZ1+0.75LR
3 1.0 6 0.75 4 0.75
LOAD COMB 18 0.6D+WX1
```

3 0.6 5 1.0
LOAD COMB 19 0.6D+WZ1
3 0.6 6 1.0
LOAD COMB 20 0.6D+0.7EX
3 0.6 2 0.7
LOAD COMB 21 0.6D+0.7EZ
3 0.6 1 0.7
LOAD COMB 22 D+WX1
3 1.0 5 1.0
LOAD COMB 23 D+WZ1
3 1.0 6 1.0
LOAD COMB 24 1.4DL
3 1.4
LOAD COMB 25 1.4D+1.4T+0.5LR
3 1.4 9 1.4 4 0.5
LOAD COMB 26 1.4D+1.4T+0.5LR
3 1.4 9 1.4 4 0.5
LOAD COMB 27 1.2D+0.8WX1+1.6LR
3 1.2 5 0.8 4 1.6
LOAD COMB 28 1.2D+1.6WZ1+0.5LR
3 1.2 4 0.5 6 1.6
LOAD COMB 29 1.2D+1.6WX1+0.5LR
3 1.2 4 0.5 5 1.6
LOAD COMB 30 1.2D+1.6WZ1+0.5LR
3 1.2 4 0.5 6 1.6
LOAD COMB 31 1.2D+EX
3 1.2 2 1.0
LOAD COMB 32 1.2D+EZ
3 1.2 1 1.0
LOAD COMB 33 0.9D+1.6WX1
3 0.9 5 1.6
LOAD COMB 34 0.9D+1.6WZ1
3 0.9 6 1.6
LOAD COMB 35 0.9D+EZ
3 0.9 1 1.0
LOAD COMB 36 0.9D+EX
3 0.9 2 1.0
LOAD COMB 39 D+0.75WX2+0.75LR
3 1.0 7 0.75 4 0.75
LOAD COMB 40 D+0.75WZ2+0.75LR
3 1.0 8 0.75 4 0.75
LOAD COMB 41 0.6D+WX2
3 0.6 7 1.0
LOAD COMB 42 0.6D+WZ2
3 0.6 8 1.0
LOAD COMB 45 D+WX2
3 1.0 7 1.0
LOAD COMB 46 D+WZ2
3 1.0 8 1.0
LOAD COMB 47 1.2D+0.8WX2+1.6LR
3 1.2 7 0.8 4 1.6
LOAD COMB 48 1.2D+1.6WZ2+0.5LR
3 1.2 8 1.6 4 0.5
LOAD COMB 49 1.2D+1.6WX2+0.5LR

3 1.2 7 1.6 4 0.5
LOAD COMB 53 0.9D+1.6WX2
3 0.9 7 1.6
LOAD COMB 54 0.9D+1.6WZ2
3 0.9 8 1.6
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
FC 28000 ALL
FYMAIN 414000 ALL
MINMAIN 16 MEMB 33 TO 36 47 50
MINMAIN 25 MEMB 1 2 5 8 18 20 22 24
MINSEC 10 ALL
DESIGN BEAM 33 TO 42 44 45 47 48 50 51 53 TO 55
DESIGN COLUMN 1 2 5 8 18 20 22 24 49 52
END CONCRETE DESIGN
FINISH

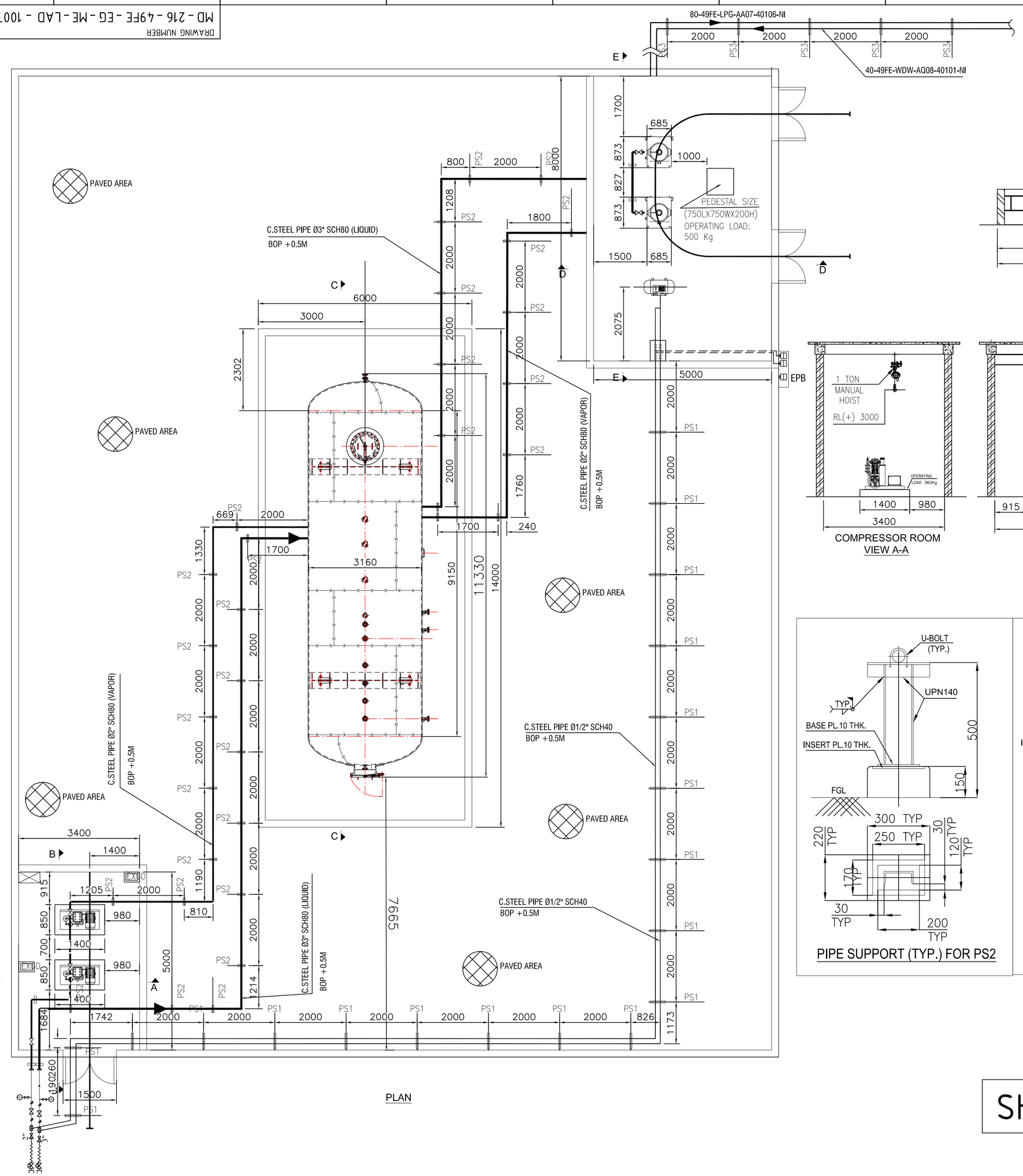
ATTACHMENT 3 LAYOUT DRAWING

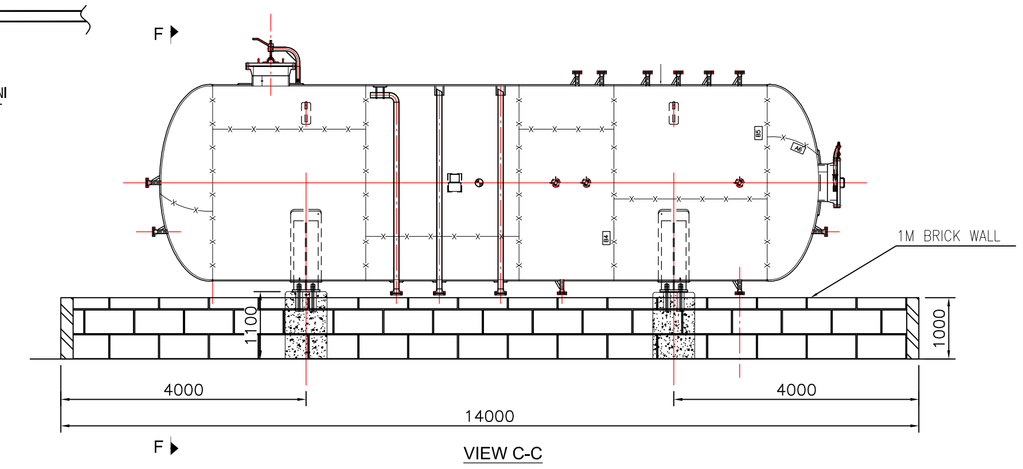
PROPRIETARY & CONFIDENTIAL

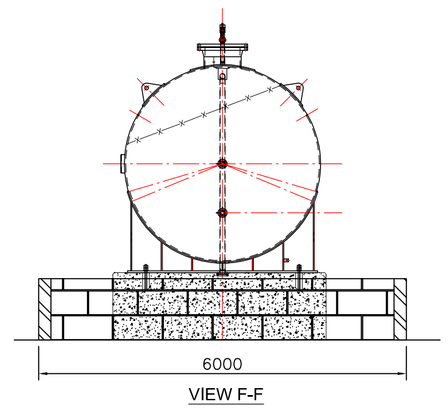
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

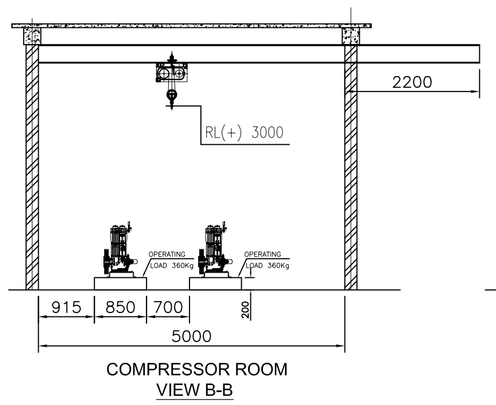
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

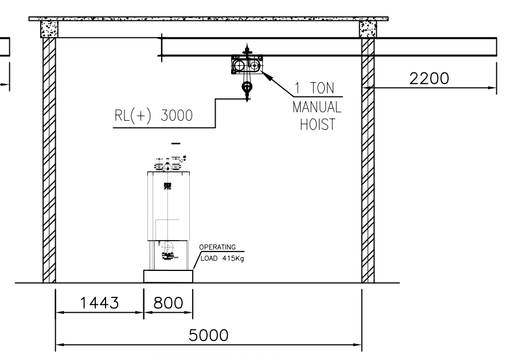
DRAWING NUMBER MD-216-49FE-EG-ME-LAD-1007 REV A01

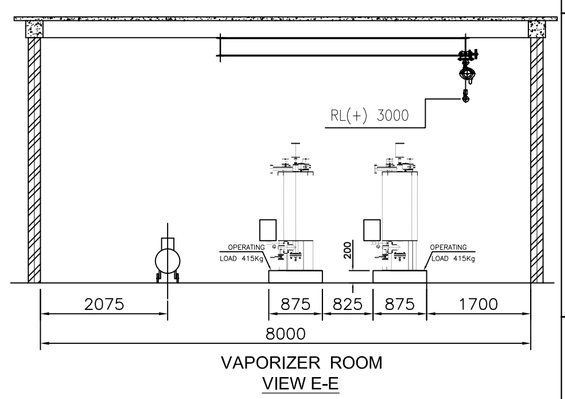


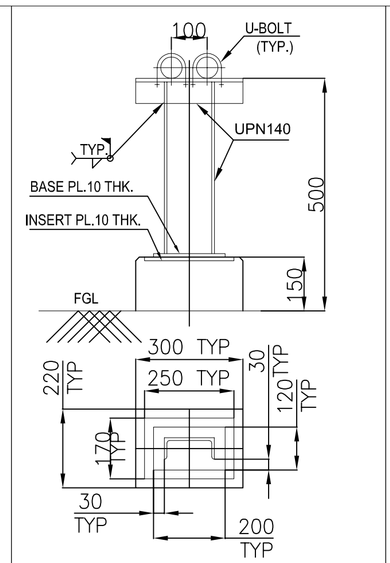


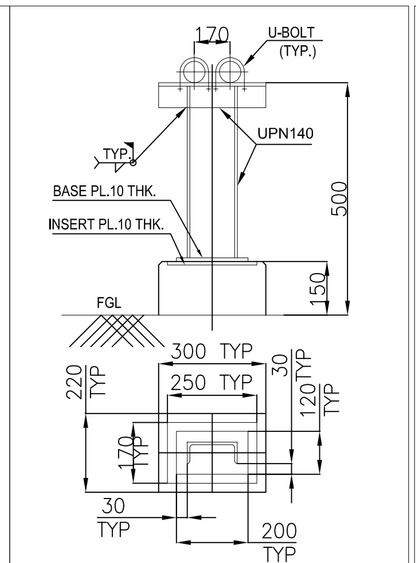












PIPE LOAD WITH WATER

S.NO.	PIPE DIA NB	PIPE WEIGHT WITH WATER KG/M
1.	15	1.2
2.	20	2
3.	25	3
4.	40	5.3
5.	50	8
6.	65	12
7.	80	16

NOTE :
1. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN M UNLESS OTHERWISE SPECIFIED.
2. 0.00 LEVEL CORRESPONDS TO FINISHED GROUND LEVEL.
3. DIMENSIONS AND SUPPORTS LOCATION WHICH EVER IS REQUIRED IF NECESSARY BESIDES THE SUPPORTS INDICATED IN DRAWING MAY BE DECIDED SUITABLY AT SITE.
4. PIPE OPENING TO BE PROVIDED WITH PUDDLE FLANGE.
5. REFER DRAWING: LPG UNLOADING, STORAGE & TRANSFER SYSTEM.
6. REFER DRAWING: LPG UNLOADING, STORAGE & TRANSFER SYSTEM - COMPRESSOR AND MOTOR SKID GA DRAWINGS, VAPORIZER GA DRAWINGS LPG BULLET GA DRAWING.

SHEET-2

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

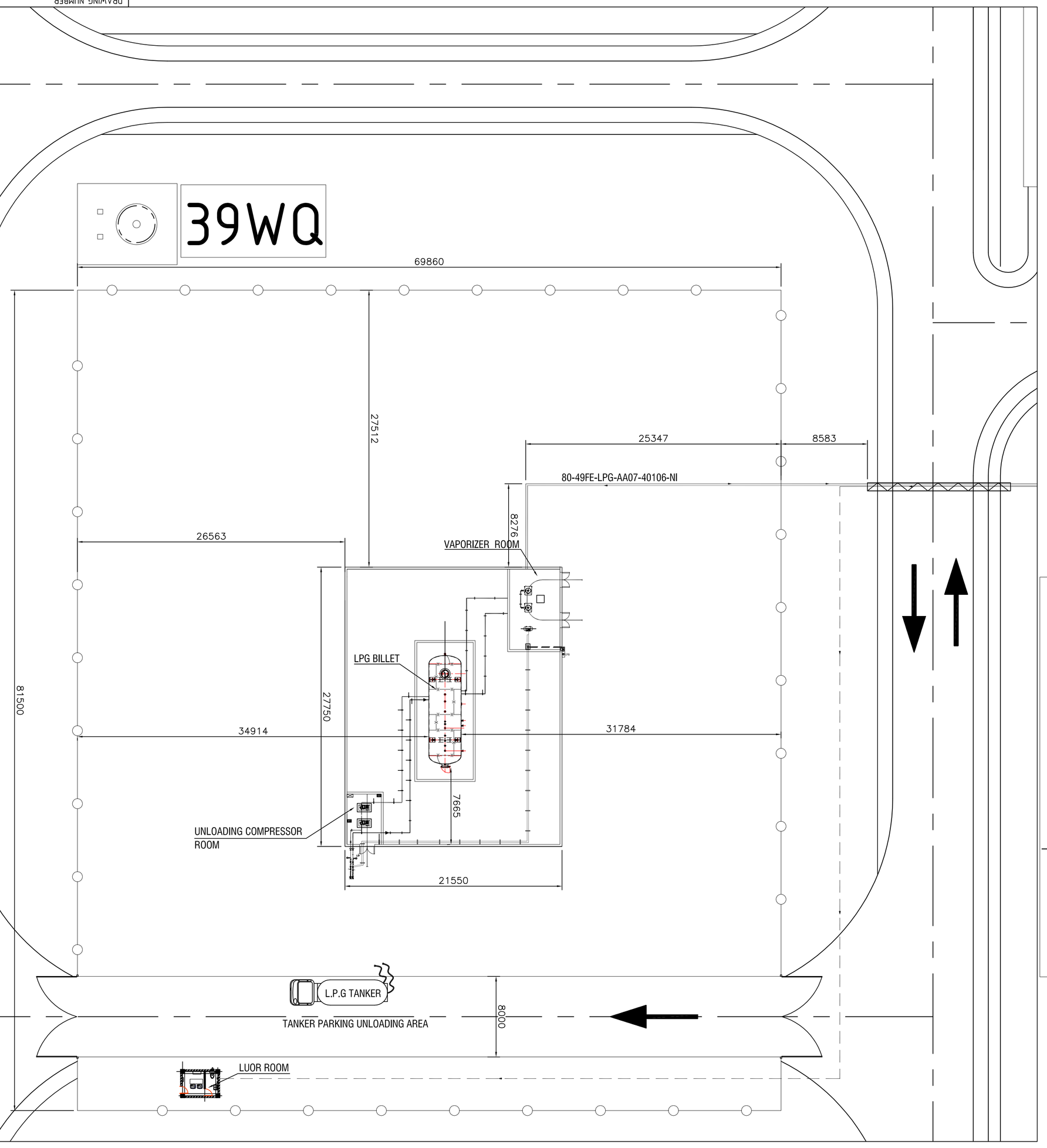
LARSEN & TOUBRO DRAWING NUMBER	019045_080_U_V_19143_007
REVISION	0
OUTOTEC DRAWING NUMBER	900707__GAU01__AG50__CLD01__49001
DOCUMENT ID	OU100694283
REVISION	0
DRAWN	
CHECKED	
ENGINEER	
APPROVED	
EMPLOYER	
SCALE	NTS
	A1

MANSOURAH - MASSARAH GOLD PROJECT
LAYOUT
FUEL OIL STORAGE
EQUIPMENT ARGT. WITH PIPING FOR LPG SYSTEM

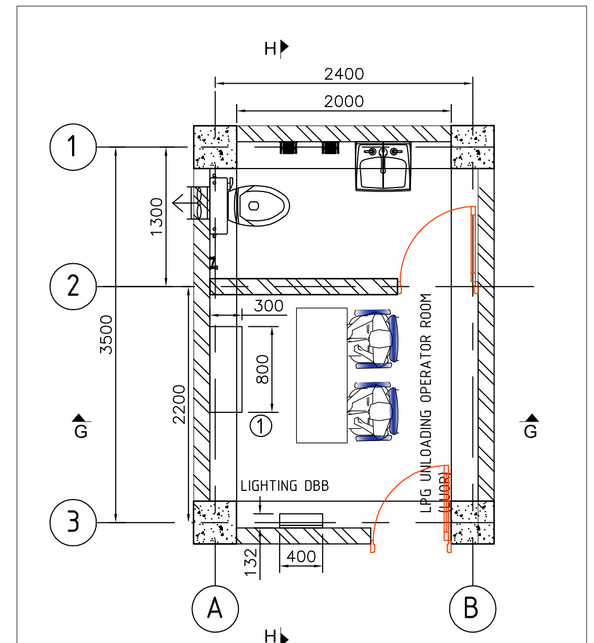
DRAWING NUMBER MD-216-49FE-EG-ME-LAD-1007 REV A01

DRAWING NUMBER MD-216-49FE-EG-ME-LAD-1007 REV A01

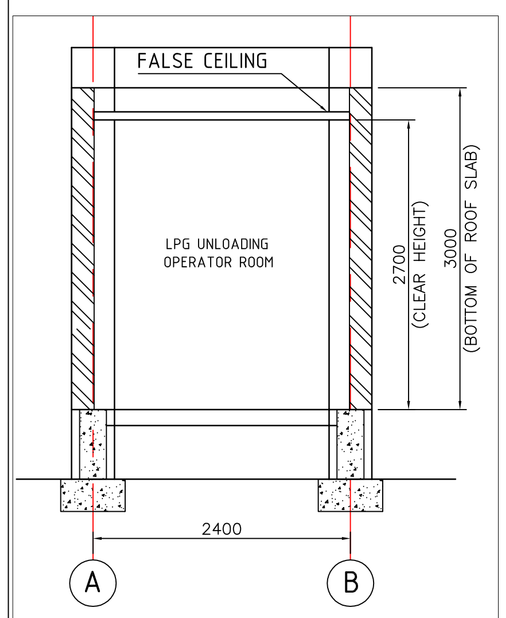
39WQ



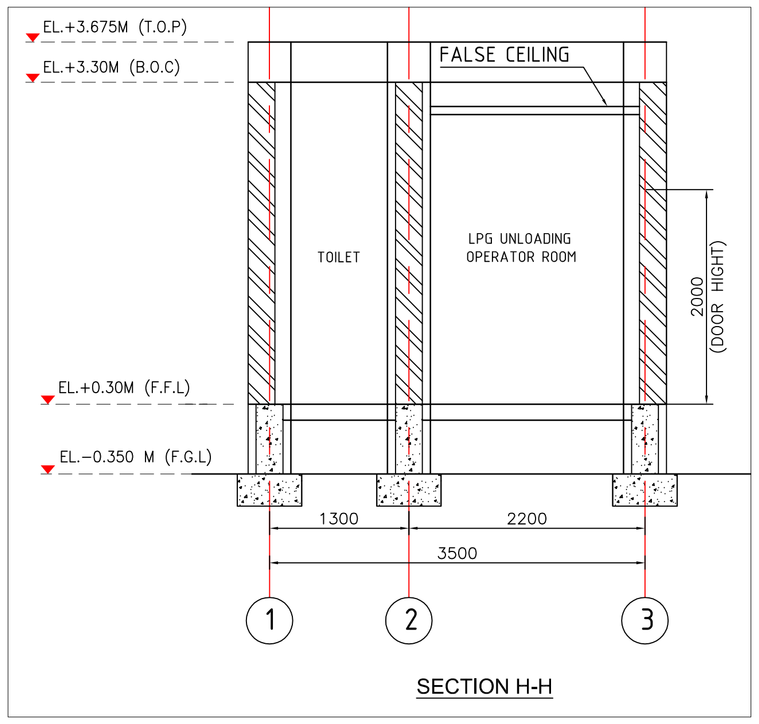
PLAN



LUOR PLAN



SECTION G-G



SECTION H-H

SHEET-1

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

LARSEN & TOUBRO DRAWING NUMBER	O19045_080_U_V_19143_007
REVISION	0
OUTOTEC DRAWING NUMBER	900707__GAU01__AG50__CLD01__49001
DOCUMENT ID	OU100694283
REVISION	0
DRAWN	
CHECKED	
ENGINEER	
APPROVED	
EMPLOYER	
SCALE	NTS
	A1

MANSOURAH - MASSARAH GOLD PROJECT
LAYOUT
FUEL OIL STORAGE
EQUIPMENT ARGT. WITH PIPING FOR LPG SYSTEM

DRAWING NUMBER MD-216-49FE-EG-ME-LAD-1007 REV A01